

Half a Century of Risk Premia in a Generalized Multi-Currency Carry Strategy: An Empirical Investigation

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Abstract

We study empirically the excess returns from a generalization of the traditional interest-rate and FX carry strategies in which the domestic, funding and investment currencies can all be different – a situation of great practical relevance. We express these excess returns in terms of economically interpretable covariances between forwards/futures ratio and the domestic stochastic discount factor. We find that these excess returns are predictable, and we make use of this predictability to create fully-out-of-sample state-dependent generalized strategies that display much higher Sharpe ratios, and much lower drawdowns, than the unconditional ones. We show that these conditional strategies benefit importantly from the ability to invest and fund in currencies different from the domestic one.

Keywords: International Carry, Expectation Hypothesis, Uncovered Interest Parity, Excess Returns, Yield Curve, Treasury Bonds

1 Introduction

The strong Interest Rate Expectation Hypothesis (IREH) states that forward bond prices are unbiased expectations of the future bond prices. Similarly, the strong FX Expectation Hypothesis (FXEH) claims that forward FX rates are unbiased expectations of future FX rates.¹ Whether the IREH and the FXEH are satisfied in the data is of obvious importance, both from the practical and the theoretical point of view. From the applications side, predictability of returns can give rise to profitable trading strategies; more generally, estimating a time-dependent risk premium gives investors an indication of when one is well or poorly rewarded for assuming duration or foreign exchange (FX) risk. From a theoretical perspective, the validity of otherwise of the two EHs tells us something fundamental

¹Following the convention in Froot (1989), we define the strong expectation as the hypothesis that forward (FX or interest) rate are unbiased predictors of future rates. The weak expectation hypothesis can be stated by saying that forward rates are biased estimators of future rates, but that this bias is not state-dependent (is a constant). In our work, unless otherwise stated, when we refer to the expectation hypothesis, we refer to the strong version.

about the covariance between the stochastic discount factor (consumption growth) on the one hand, and the payoffs of fixed income securities or FX rates. In a rational asset pricing context, deviations from the expectation hypothesis also gives us information about preferences, particularly regarding the marginal utility's rate of decline with increasing wealth, and about investors' dislike for uneven consumption.

So-called carry strategies are usually used in empirical studies to ascertain whether the EHs hold. Carry strategies, both in interest rates and in FX, consist of funding an investment strategy by issuing a bond in one currency, investing the proceeds in a longer- or same-maturity bond in the same or a different currency, unwinding the long and short positions, and recording the positive or negative profit. The traditional interest rate carry strategy involves issuing a short-term bond and investing in a longer-term bond within the same currency. These are the strategies underpinning the analyses of Campbell and Shiller (1991), Fama and Bliss (1987), Cochrane and Piazzesi (2005), Cieslak and Povala (2015), and many others. Conversely, the traditional FX carry strategy, or Uncovered Interest Rate Parity (UIP) carry strategy, involves bonds of the same maturity but in different currencies (see, eg, Ang and Chen (2010a); Bekaert, Wei, and Xing (2007); Pesch, Piatti, and Whelan (2023)).

A moment's reflection shows that the traditional interest rate carry and the UIP strategies are just special cases of a more general strategy: an investor in currency X could generally fund herself in a currency L (which may, but not necessarily coincide with X) and invest in a bond denominated in currency H (which, again, may differ from both X and L), of maturity equal to or longer than the investment horizon. (With our notation, the traditional interest-rate carry strategy is recovered when currency $X = H = L$ and when the funding bond has a shorter maturity than the investment bond; the UIP strategy corresponds to the case when the maturities of the funding and investment bonds are the

same, and $L \neq H$.)² In the highly integrated international financial markets, investors (hedge funds, asset managers, trading desks) often have geographically unconstrained trading mandates. So, a, say, British bank may well engage in the very common \$/Yen carry strategy. Whatever the profit of this strategy, at expiry the British bank will record its profits in sterling pounds, not in US \$ or Japanese yen, and will have therefore engaged in the generalized carry strategy that we analyze in this work. In general, whether the investors buys a bond of maturity longer than or equal to the maturity of the currency-mismatched investment bond depends on whether she has views, not only about the exchange rate between the funding and investment currency, but also about the behaviour of rates in the investment currency. In general, we see no reason why the trading views should be ‘one-dimensional’, especially given the integrated approach followed by sophisticated players such as hedge funds or fixed-income asset managers.³ So, we would argue that the most general three-currencies strategies, in which funding, investments and profits are potentially all in different currencies, are the financially most relevant ones. We use half a century of joint realizations of FX rates and bond prices in four major currencies to explore what this generalized carry strategy can tell us about risk compensation and cross-currency trading opportunities.

Despite its appeal, to our knowledge this more general strategy has rarely been analyzed in the literature. Apart from missing an important investment dimension, this neglect has also been unfortunate from the theoretical perspective, because looking at the more general three-currency strategy provides useful insights that the special cases do not provide.

²A comment on notation. Unless otherwise stated, we denote by H the currency of investment, L the currency of funding, and by X the currency of the investor. (The choice of symbols stems from the preference by carry traders to invest in the *H*igh-yielding while funding themselves in the *L*ow-yielding currency. It is this high-low feature that then justifies the (positive) ‘carry’ moniker for the strategy. As for the designation of the currencies, we denote dollars, sterling, euros and yen by USD, GBP, EUR and JPN, respectively.

³For instance, at the largest fixed-income asset manager in the world, PIMCO, there exist specialized desks with large trading mandates that are allowed to invest in and use for funding bonds of any currency and any maturity.

This can be seen as follows. If an agent invests in a bond with maturity greater than her same-currency funding loan, she implicitly expresses a view on the future yields in that currency. If she invests in a bond denominated in a currency other than the same-maturity funding loan, she implicitly expresses a view on future FX rates. If she combines a maturity mismatch with a currency mismatch, she implicitly expresses a view not only on future yields and future exchange rates, but also *on the correlation between the two*. By combining aspects of investing in long-term maturity bonds (characteristic of the traditional interest-rate carry) and funding or investing in a different currency (a feature of the traditional uncovered interest rate parity (UIP) strategy), the more general carry strategy introduces an extra source of return (risk compensation) that can be important. Interestingly, we find that the risk premium of the more general strategy does not simply equal the sum of the risk premia of the two traditional carry strategies. We show that this extra term (this extra risk premium) arises from the difference between two key covariances: the covariance of domestic consumption growth with domestic bond values, and the covariance of domestic consumption growth with investment bond values. And, indeed, we find that the profitability that can be reaped from the more general strategy is different from the sum of the profitabilities of the individual ('stand-alone') strategies.

In brief our results can be summarized as follows. First, we show how the expected excess returns from each strategy can be decomposed as a combination of three identifiable and interpretable risk premium contributions (covariances between forward/futures ratios and stochastic discount factors), and we show how these contributions can be estimated. We show that the risk premium associated with bearing interest-rate ('duration') risk is the dominant contributor to profitability. The covariance between the stochastic discount factor and the FX forward/futures ratio contributes positively in the investment leg and negatively in the funding leg, but the investment leg dominates and the overall contribution from this term is therefore positive. For the long-always ('unconditional') strategy, the benefit of the extra flexibility afforded by choosing an investment or funding currency different

from the domestic one is negated by increase in volatility coming from the currency mismatch. However, we find that the excess return process is predictable, and well described by an Ornstein-Uhlenbeck (AR(1)) process. Thanks to this predictability we construct a set of fully out-of-sample conditional strategies, and we show that their Sharpe Ratio is statistically different and economically superior to the unconditional strategies, and is also superior to the strategy often employed by market professionals of systematically investing in the highest-yielding and funding in the lowest-yielding currency.

In view of the above, the paper is structured as follows: in Sec 2 we present a literature review; the data are described in Section 3; in Section 4 we present the investigation strategy that allows us to decompose the excess returns into their risk premia components, and to identify the latter; the empirical results are presented, discussed and interpreted in Section 5, for the unconditional case first (Section 5.1), and then for the conditional case (Section 5.4). Finally, Section 6 presents our conclusions, and suggestions for further work. In Appendix A we describe the resampling procedure that we have used in the statistical analysis, and present the tests that we have run to validate its applicability to the problem at hand.

2 Literature Review

Our paper is linked to the Interest Rate Expectation Hypothesis (IREH) and Uncovered Interest Rate Parity (UIP) literature. Since the pioneering work by Fama and Bliss (1987) and Campbell and Shiller (1991), same-currency carry strategies have been used as empirical diagnostic tools to understand whether the IREH holds, and whether return-predicting factors exist that can reliably predict fixed-income excess returns. The same-currency return-predicting literature is vast (see Rebonato (2018) for a critical review), and, since the work by Cochrane and Piazzesi (2005) in the early 2000s, it has seen a veritable renaissance, with the explanatory variables chosen to be yield curve variables (as in Cochrane and Piazzesi (2005)), macroeconomic variables (as in Ludvigson and Ng (2009)), or a mixture of the

two (as in Cieslak and Povala (2015)). Overall, the evidence against the IREH is largely consistent, with the literature almost unanimously rejecting the null hypothesis (for a by-now-rather-dated dissenting voice, see Froot (1989)).

Similarly vast is the literature on violations of the UIP (see, eg, Koijen, Moskowitz, Pedersen, and Vrugt (2018) and MacDonald and Nagayasu (2015)). The UIP dominates the discussion on exchange rate determination in international finance, and there appears to be overwhelming empirical evidence against UIP at least for holding periods less than one year (Bilson (1981); Fama (1984); Bekaert and Hodrick (1993); Engel (1996); Froot and Thaler (1990); Mark and Wu (1998); Chaboud and Wright (2005); Berg and Mark (2019)). The cross-currency carry trade has been studied by considering two types of strategies. The first is the traditional carry strategy that goes long (short) the same-maturity bonds of high (low) interest-rate currencies. The second strategy is the slope carry that goes long (short) the long-term bonds of steep (flat) yield curve currencies. Andrews, Colacito, Croce, and Gavazzoni (2024) studied these two strategies and found empirical evidence that slope carry strategies had negative returns before 2008 but significant positive returns after the 2008 crisis while traditional carry returns were lower after 2008. They use a DSGE model and attribute the returns accompanied by steeper yield curves and low interest rates to high exposure to global inflation in the post-2008 period.

The first original feature of our study is that, while most studies of the UIP focus on the traditional carry strategy of purchasing high-interest rate currencies and selling low-interest rate currencies with the same horizon, we investigate a strategy where the maturity of funding bonds can differ from the maturity of the investment periods. So, for instance, Ang and Chen (2010a) study the carry strategy with both the fund and interest rate of one month. Andrews, Colacito, Croce, and Gavazzoni (2024) studied the traditional carry strategy with both the fund and interest rate of three months. Although Bekaert, Wei, and Xing (2007), Lustig, Stathopoulos, and Verdelhan (2019), and Pesch, Piatti, and Whelan (2023) study

the traditional UIP strategy at long horizons, the funding bonds in their strategies have the same long horizon as the investment bonds. In our strategy, the investment bond, which can be in a different currency from the funding bond, does not come to maturity by the end of the investment-holding periods (12 months in our case). As we shall see, this is important, because it introduces a correlation between FX and investment yields – a correlation that vanishes in traditional UIP strategies where both the investment and funding bonds mature after the investment horizon and is absent in same-currency bond carry strategies. This correlation (covariance) gives an additional potential source of risk premium, that cannot be captured with the traditional UIP carry strategy. So, for instance, the work by Lustig, Stathopoulos, and Verdelhan (2019), who studied long-term carry using 10-year maturity bonds both for investment and funding and found that long-term carry trade is not significantly profitable, misses this potential source of extra return.

The second distinctive feature of our study is that we study the international carry in a general framework without imposing *a priori* the investment, the funding, and the domestic currencies. Most current studies on currency carry strategies use the US as the domestic investor and only consider the two-currency (domestic-investment) combination rather than the three-currency (fund-domestic-investment) combination. We consider the richer three-currency strategies from the perspective of US, EUR, GBP and JPN investors, and express profits in their local currency. Pesch, Piatti, and Whelan (2023) study the strategy that goes long foreign long-term bonds and short a long-term US bond. Berg and Mark (2019) also study a two-country DSGE model to explore international currency returns rather than the three-currencies (domestic funding and investment). Thus, our work enjoys greater generality, is more realistic (as mentioned, trading desks, hedge funds and portfolio managers often have unconstrained trading mandates), and is of relevance to investors who may not be \$-denominated. The research of Lustig, Stathopoulos, and Verdelhan (2019) is similar to our work because they studied the excess return between the return of investing long-term foreign bonds and the domestic short-term risk-free rate,

but they also only consider the two-currency pair from the sole perspective of US investors and set USD as the domestic currency. Similarly, Andrews, Colacito, Croce, and Gavazzoni (2024) study both the traditional carry and the slope carry, but only from the perspective of US investors, and Ang and Chen (2010a) try to predict the traditional carry returns using a number of return-predicting factors for various two-currency pairs among G10 countries. In sum: all these authors work in a two-country framework; do not considering funding in a different currency than the domestic currency; and measure all the returns in USD—even for strategies where domestic investors are not US-based.

The third salient feature of our study is that we focus both on explaining the different risk premia associated with our strategy and on finding a good return predicting factor (RPF). A lot of academic work has been devoted to testing the predictive power of various RPFs for international carry strategies. The difference in short rates across countries is the most widely used RPF for FX carry strategies. Ang and Chen (2010a) and Campbell and Shiller (1991) studied the currency carry strategy that funds and invests in same-maturity bonds in different currencies. They find that other yield curve predictors like changes in interest rates and term spreads significantly predict excess foreign exchange returns. Currency portfolio returns based on changes in interest rates and term spreads exhibit low skewness risk and have a low correlation with carry returns. Predictability from these yield curve variables persists for up to 12 months and is robust to controlling for other predictors of currency returns. However, for long-term traditional carry strategies, it has proven difficult to predict their returns only based on interest-rate information. Lustig, Stathopoulos, and Verdelhan (2019) studied both the traditional and the slope currency carry strategies and found that as the maturity of bonds increased, the average excess return declined to zero, indicating that the power of the slope factor to predict cross-country differences in dollar bond returns disappears. Ang and Chen (2010a) find that the predictive power of their RPFs persists up to 12 months. In the context of multi-currency strategies, Feng (2023) tried to use Cochrane-Piazzesi factor for the investment leg of the strategy, but found lim-

ited predictive power, especially for long-term investments (R^2 value less than 0.1). Bekaert, Wei, and Xing (2007) also finds that, when long-term horizons are considered, the deviation from UIP is limited.

Another strand of analysis typically begins with the theoretical estimation of the risk premia of the carry strategy based on different interest rate models; then the predicting power of the RPFs suggested by various interest rate models are explored. In this strand, studies by Backus, Foresi, and Telmer (2001), Ahn (2004), and Brandt and Santa-Clara (2002) are based on multi-factor interest rate models. Ang and Chen (2010a) uses a simple one-factor model with stochastic volatility to model short-term yields. And several authors, such as Brennan, Wang, and Xia (2004) and Lettau and Wachter (2011), have modelled the risk factor itself to be a time-varying latent process. Although our study also starts from the decomposition of the risk premia of a general carry strategy, we do not specify a model of interest rates. Rather, we find that the excess-return process displays strong mean-reverting properties, and that this makes it predictable. We show that fully-out-of-sample strategies that exploits this predictability displays very attractive Sharpe ratios – Sharpe Ratios statistically and economically higher than the corresponding unconditional (long-always) strategies.

Finally, we note in closing this section a theoretically minor, but practically significant, source of difference between our study and the existing literature: we estimate returns by calculating explicit changes in bond prices, while most of the literature uses linearization and derives return results in terms of yields (see, eg, Pesch, Piatti, and Whelan (2023); Ang and Chen (2010a); Andrews, Colacito, Croce, and Gavazzoni (2024); Bekaert, Wei, and Xing (2007)). For short maturities, this approach yields almost the same results, but when maturities extend to 10 years, we find that the yield approximation can lead to non-trivial differences (because of convexity effects).

3 Data Description

Our empirical analysis focusses on Treasury (government) bonds issued in four major international currencies: US dollars, British pounds, Euros and Japanese yens, referred to in what follows as USD, GBP, EUR, JPN. The USD Treasury yield data are sourced from the widely used Gürkaynak, Sack, and Wright (2007) source, updated to cover data to September 2022. German yield time series data, denominated in EUR, are obtained from the Bundesbank database.⁴ GBP yield time series are extracted from the Bank of England database, while Japanese bond data are sourced from the Japanese Ministry of Finance statistics website. As for spot foreign exchange (FX) rates, our data are sourced from the Federal Reserve Economic Data (FRED) website, maintained by the St Louis Fed. Fig 2 plots the time series of GBP, EUR, and JPN exchange rates against the USD. For our analysis, we construct cross-currency carry strategies based on 12 currency pairs – EUR/USD, GBP/USD, JPN/USD, USD/EUR, GBP/EUR, JPN/EUR, USD/GBP, EUR/GBP, JPN/GBP, USD/JPN, EUR/JPN, GBP/JPN. The last nine (‘cross’) FX rates are calculated from the first three FX rates with USD as base currency.

We collect the yields of government bonds, inflation, and FX data for each of these four countries at a monthly frequency. Since currency exchange rates were fixed under the Bretton Woods system, we begin our sample in September 1974, which is well after the breakdown of Bretton Woods in the early 1970s, and end in September 2022. Our sample represents approximately 600 months (roughly, half a century) in total. We calculate the time- t bond price of a discount bond of residual maturity T , P_t^T , from the zero-coupon

⁴For periods predating the introduction of the EUR currency, German Bund prices are utilized. Since Germany has the largest GDP among the eurozone countries, we consider the German Deutsche Mark to be representative of the euro prior to its introduction in December 1998. This assumption has been widely used in the literature (see, eg, Andrew, Colacito, Croce, and Gavazzoni (2024), Ang and Chen (2010b) and Lustig, Roussanov, and Verdelhan (2011)).

	USD	GBP	EUR	JPN	\$/£	\$/EUR	\$/Yen
USD	1	0.928	0.888	0.862	-0.632	0.352	-0.784
GBP	0.928	1	0.946	0.935	-0.570	0.490	-0.810
EUR	0.888	0.946	1	0.871	-0.473	0.492	-0.685
JPN	0.862	0.935	0.871	1	-0.610	0.485	-0.853
\$/£	-0.632	-0.570	-0.473	-0.610	1	0.070	0.808
\$/EUR	0.352	0.490	0.492	0.485	0.070	1	-0.361
\$/Yen	-0.784	-0.810	-0.685	-0.853	0.808	-0.361	1

Table 1: The correlation between the average yields in the four currencies and the exchange rates against the dollar (levels).

bond yield, y_t^T , through the formula in Eq 1:

$$P_t^T = e^{-y_t^T \cdot T} \quad (1)$$

Fig 1 shows the time series of the average yield for US, EU, GBP, and JPN. For each country, the average yield is calculated as the arithmetic mean of the yields on the country's treasury bonds with maturities ranging from 1 to 9 years. Fig 2 then shows the cost in \$ of one unit of GBP, EUR and JPN currency (in the latter case the exchange rate has been multiplied by 100 to facilitate visual comparison). The yield graph shows a high degree of comovement among different-currency yields. This is quantified in Tab 1, which shows the correlation between the average yields in the four currencies and the exchange rates against the dollar (levels). Tab 2 presents the same information for percentage differences. We will discuss the significance of these empirical findings when we examine the profitability of our carry strategy.

	USD	GBP	EUR	JPN	\$/£	\$/EUR	\$/Yen
USD	1	0.676	0.477	-0.205	0.067	-0.038	-0.081
GBP	0.676	1	0.328	-0.112	0.025	0.031	-0.041
EUR	0.477	0.328	1	-0.249	-0.099	-0.162	-0.106
JPN	-0.205	-0.112	-0.249	1	0.314	0.498	0.279
\$/£	0.067	0.025	-0.099	0.314	1	0.434	0.233
\$/EUR	-0.038	0.031	-0.162	0.498	0.434	1	0.431
\$/Yen	-0.081	-0.041	-0.106	0.279	0.233	0.431	1

Table 2: The correlation between the percentage differences in average yields in the four currencies and percentage differences the exchange rates against the dollar.

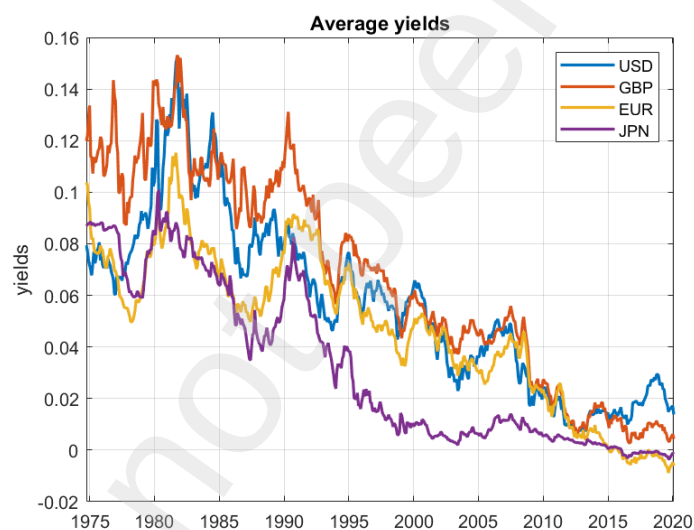


Figure 1: Time series of the average yield for US, EU,GBP, and JPN. For each country, the average yield is calculated as the arithmetic mean of the yields on the country's treasury bonds with maturities ranging from 1 to 9 years.

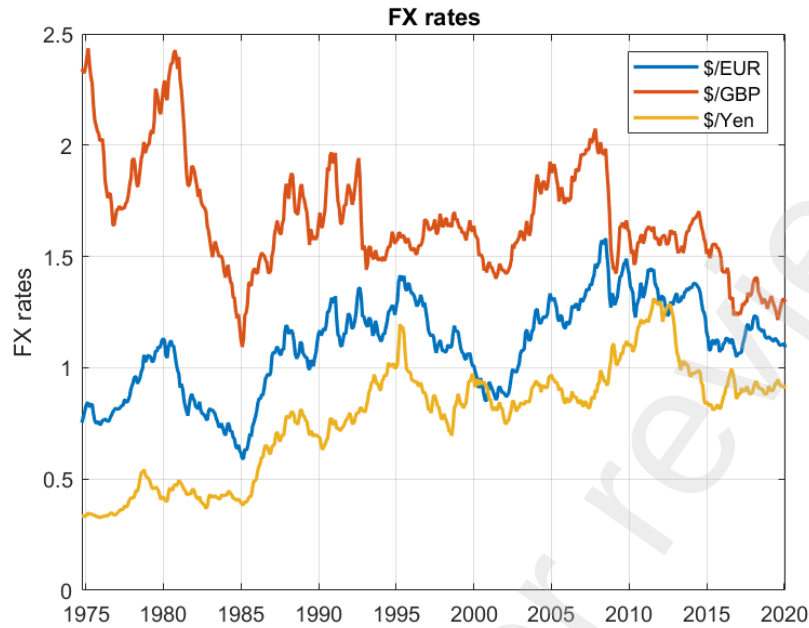


Figure 2: Time series of the foreign exchange rate of EUR, GBP, and JPN against USD dollar. The rate $\$/Yen$ has been multiplied by 100 for ease of comparability.

For the moment we simply draw the attention to the entries in Tab 2 that we have highlighted in red. These entries are noteworthy because they show i) a negative correlation between JPN yield changes and the yields of all other currencies (while all the other yield-change correlations are positive); and ii) a strongly significant and positive correlations between JPN yields and the all the $\frac{\$}{A}$ FX rates (where A is any non-\$ currency), while all the other yield/FX correlations are small and, in general, not statistically significant. When it comes to explaining changes in yields and FX rates, there are, of course, a multitude of macrofinancial factors at play. It is, however, at least suggestive to note that the Japanese Yen has been the currency of choice for the so-called ‘High-Low’ carry strategy, in which the purchase of a high-yielding bond is financed by the sale of the lowest-yielding bond (we discuss this strategy in some detail in Section 5.4). Since, as Fig 2, JPN yields have been the lowest for most of the time in our sample, this would mean shorting JPN bonds and buying higher-yielding Treasury bonds. *Ceteris paribus*, by itself this would create a

negative correlation between JPN bonds and the higher-yielding bonds (as observed), and a positive correlation between JPN yields and the various $\frac{\$}{A}$ FX rates (also as observed). We do not pursue this line of inquiry further, but leave it for future analysis.

4 The Investigation Methodology

4.1 Notation and Definitions

In this study, the international generalized carry strategy is to borrow money in one currency by shorting 1-year maturity treasury bonds in that currency, to hold the position to maturity of the funding bond, and to invest in another currency by buying long-term (2-10 years maturity) treasury bonds of (possibly) another country, and, if needed, to covert the proceeds in the currency of the investor. The symbols used in our analysis are defined in the Table 3:⁵

⁵Since we denote by $FX(A, B)$ the value in currency A of one unit of currency B , for future reference we recall that the following obvious relationships hold

1. $FX(A, B) \cdot FX(B, C) = FX(A, C)$
2. $FX(A, B) = \frac{1}{FX(B, A)}$
3. $\frac{FX(A, B)}{FX(A, C)} = FX(C, B)$
4. $FX(A, A) \equiv 1$

Term	Explanation
X	The domestic currency (currency of the investor).
H	The investment currency in which bonds are bought.
L	The funding currency in which bonds are issued.
PL_t^T	The value of zero-coupon bond issued in currency L at time t with maturity T .
PH_t^T	The value of zero-coupon bond issued in currency H at time t with maturity T .
$FX_t(A, B)$	The value rate of one unit of currency A per unit of B at time t : $\frac{A}{B}$; A and B can be any pair of currencies.
$FFX_0(A, B; T)$	The time-0 forward foreign exchange rate of one unit of B in currency A for delivery at T. $FFX_0(A, B; T) = FX_0(A, B) \cdot \frac{PB_0^T}{PA_0^T}$.
$FPA_0^n(\tau)$	The time-0 forward bond price in currency A for delivery τ ; A can represent the currency of X, L, or H. $FPA_0^n(\tau) = \frac{PA_0^n}{PA_0^T}$.
Π_t	Profit and loss of a portfolio strategy; t can be 0 or 1 in our study.
$M_0^1(X)$	The nominal stochastic discount factor (SDF) that can discount pay-off at time 1 to time 0 for an agent using currency X.
$g_c(X)$	The domestic consumption growth rate.
ϵ^n	The difference between the excess return of our strategy and the sum of two traditional carries - traditional FX/UIP carry and traditional interest rate carry. The n is the maturity of investment bonds.
$\mathbb{E}[\cdot]$	The expectation operator.
$\text{COV}[\cdot]$	The covariance operator.

Table 3: The definition of quantities used in the study.

4.2 Construction of the Cross-Currency Carry Strategy

The generalized carry strategy for an investor with domestic currency X consists of using a fixed (unit) amount of currency X to invest in a long-dated bond in currency H , while funding the position for 12 months in currency L . The strategy is held for 12 months, at which point, positions are liquidated, and all debts are repaid. Each month, the investor invests the same (unit) amount of currency X . So, at any point in time, there are 12 active strategies – each started one month apart. The returns are therefore going to be overlapping.⁶

The construction of this strategy can be decomposed in the following three steps.

Step 1 – Funding. First, at time 0, the X -currency investor borrows $\frac{1}{FX_0(X,L)}$ units of currency L . 1 unit of currency L can be obtained by issuing an amount $\frac{1}{PL_0^1}$ of the 1-period bond in currency L . So, the position in the bond in currency L requires for funding $\frac{1}{FX_0(X,L)}$ units of currency L is $-\left[\frac{1}{FX_0(X,L)} \frac{1}{PL_0^1}\right]$. This amount of borrowing is equivalent to $\left[\frac{1}{FX_0(X,L)} \frac{1}{PL_0^1}\right] \cdot PL_0^1 \cdot FX_0(X,L) = 1$ unit of domestic currency X and can be converted to $\frac{1}{FX_0(X,H)}$ units of currency H .

Step 2 – Investing. Second, at time 0, we invest this amount of currency H in a long-dated bond. The investment notional is $+\left[\frac{1}{FX_0(X,H)} \frac{1}{PH_0^n}\right]$. Therefore, at time 0, the value of the portfolio, Π_0 , in currency X is given by Eq.2

$$\Pi_0 = -\underbrace{\left[\frac{1}{FX_0(X,L)} \frac{1}{PL_0^1}\right] \cdot PL_0^1 \cdot FX_0(X,L)}_{\text{L notional funding}} + \underbrace{\left[\frac{1}{FX_0(X,H)} \frac{1}{PH_0^n}\right] \cdot PH_0^n \cdot FX_0(X,H)}_{\text{H notional investing}} = 0 \quad (2)$$

Step 3 – Liquidation. Finally, at time 1, we close our position in Steps 1 and 2. Since $PL_1^1 = 1$, the value of the portfolio, Π_1 , in domestic currency X at time 1 is given by Eq.3

$$\Pi_1 = -\underbrace{\left[\frac{1}{FX_0(X,L)} \frac{1}{PL_0^1}\right] \cdot 1 \cdot FX_1(X,L)}_{\text{L notional funding}} + \underbrace{\left[\frac{1}{FX_0(X,H)} \frac{1}{PH_0^n}\right] \cdot PH_1^{n-1} \cdot FX_1(X,H)}_{\text{H notional investing}} \quad (3)$$

⁶When we study what we call the conditional generalized carry strategy we will use a 1-month holding and financing period. Everything else remains the same, but the trades are no longer overlapping.

Applying the definition of $FFX_0(A, B; T)$ and $FPA_0^n(\tau)$ in Tab 3 to Eq 3, the strategy's profit and loss can also be written as:

$$\Pi_1 = \frac{1}{PX_0^1} \left\{ -\frac{FX_1(X, L)}{FFX_0(X, L; 1)} + \frac{PH_1^{n-1}}{FPH_0^n(1)} \cdot \frac{FX_1(X, H)}{FFX_0(X, H; 1)} \right\} \quad (4)$$

We note that, apart from the pre-multiplying factor, $\frac{1}{PX_0^1}$, the three terms inside the curly brackets are all ratios of future quantities (FX or bond prices) to their respective forward quantity. This immediately shows that, if all future quantities were equal, on average, to the corresponding forward quantities, the expected profit of the strategy would be zero. Each separate deviation of the three ratios from unit brings about a potential source positive or negative expected profit for the strategy.

4.3 Introducing Risk Premia

One fundamental underlying principle in finance research (see, eg, Back (2010) and Cochrane (2009)) is that the appropriately discounted excess return of a strategy that requires no initial outlay must be 0. By 'appropriately discounted' we mean that all future payoffs should be discounted by the state-dependent stochastic discount factor (SDF), which, as shown in Table 3, we denote by $M_0^1(X)$. So, if x_1 is the (stochastic) payoff in currency X at time 1 of a given strategy, its price at time 0, P_0 , is given by $P_0 = \mathbb{E}[M_0^1(X) \cdot x_1]$. Therefore, for our zero-initial value, self-financing strategy, we have:

$$\mathbb{E}[\Pi_1 \cdot M_0^1(X)] = 0 \quad (5)$$

where $M_0^1(X)$ is the nominal SDF that can discount any payoff from time 1 to time 0 for an investor using currency X . Given the price of a zero-coupon bond issued in currency X at time t with maturity 1 year is PX_0^1 , we also have:

$$\mathbb{E}[M_0^1(X)] = PX_0^1 \quad (6)$$

Combining Eqs 4 and 5 we therefore obtain,

$$0 = -\frac{1}{PX_0^1} \cdot \mathbb{E} \left[\frac{FX_1(X, L)}{FFX_0(X, L; 1)} \cdot M_0^1(X) \right] + \frac{1}{PX_0^1} \cdot \mathbb{E} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)} \cdot \frac{FX_1(X, H)}{FFX_0(X, H; 1)} \cdot M_0^1(X) \right] \quad (7)$$

In general $\mathbb{E}[a \cdot b] = \mathbb{E}[a] \cdot \mathbb{E}[b] + \mathbb{COV}[a, b]$, but we note that the last term in Eq 7 contains the covariance of a product of two terms with the SDF. We therefore make use of the results by Bohrnstedt and Goldberger (1969) to decompose this term. Bohrnstedt and Goldberger (1969) in fact shows that:

$$\mathbb{COV}[a \cdot b, c] = \mathbb{E}[a] \cdot \mathbb{COV}[b, c] + \mathbb{E}[b] \cdot \mathbb{COV}[a, c] + \Delta \quad (8)$$

where Δ is a term that is identically equal to zero if the three variables are jointly normally distributed (and is otherwise usually small). Substituting the expectation of nominal SDF by PX_0^1 as shown in Eq 6 and expanding the expectation of products through methods of Bohrnstedt and Goldberger (1969), one gets:

$$0 = -\mathbb{E} \left[\frac{FX_1(X, L)}{FFX_0(X, L; 1)} \right] + \frac{1}{PX_0^1} \mathbb{COV} \left[-\frac{FX_1(X, L)}{FFX_0(X, L; 1)}, M_0^1(X) \right] + \mathbb{E} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)} \cdot \frac{FX_1(X, H)}{FFX_0(X, H; 1)} \right] + \frac{1}{PX_0^1} \mathbb{E} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)} \right] \cdot \mathbb{COV} \left[\frac{FX_1(X, H)}{FFX_0(X, H; 1)}, M_0^1(X) \right] + \frac{1}{PX_0^1} \mathbb{E} \left[\frac{FX_1(X, H)}{FFX_0(X, H; 1)} \right] \cdot \mathbb{COV} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)}, M_0^1(X) \right] + \Delta \quad (9)$$

At the end of Eq 9, the Δ term is the residual term of Bohrnstedt and Goldberger (1969) expansion. As mentioned, this term is exactly equal to zero if we assume the nominal SDF, the ratio of future-to-forward foreign exchange rate, and the ratio of future-to-forward bond price are jointly normally distributed. If the joint-normality assumption is not satisfied, this is in general not true, but we confirm in what follows that the term is small.

4.4 A New Expression for the Expected Excess Return

Given the decomposition of the risk premia derived above, we can provide a new and more insightful expression for the expected excess return of the generalized carry strategy. To do so, we proceed as follows. After rearranging Eq 9, and moving all covariance terms to the RHS, we obtain:

$$\begin{aligned}
 & -\mathbb{E}\left[\frac{FX_1(X, L)}{FFX_0(X, L; 1)}\right] + \mathbb{E}\left[\frac{PH_1^{n-1}}{FPH_0^n(1)} \cdot \frac{FX_1(X, H)}{FFX_0(X, H; 1)}\right] = \\
 & \underbrace{\frac{1}{PX_0^1}}_a \underbrace{\text{COV}\left[\frac{FX_1(X, L)}{FFX_0(X, L; 1)}, M_0^1(X)\right]}_\alpha \\
 & - \underbrace{\mathbb{E}\left[\frac{PH_1^{n-1}}{FPH_0^n(1)}\right]}_b \cdot \underbrace{\frac{1}{PX_0^1} \text{COV}\left[\frac{FX_1(X, H)}{FFX_0(X, H; 1)}, M_0^1(X)\right]}_\beta \\
 & - \underbrace{\mathbb{E}\left[\frac{FX_1(X, H)}{FFX_0(X, H; 1)}\right]}_c \cdot \underbrace{\frac{1}{PX_0^1} \text{COV}\left[\frac{PH_1^{n-1}}{FPH_0^n(1)}, M_0^1(X)\right]}_\gamma + \Delta
 \end{aligned} \tag{10}$$

The LHS of Eq 10 is the expected excess return, $\mathbb{E}[xret]$, of the cross-currency carry strategy. On the RHS of Eq 10, the first and second covariance terms are the risk premia from the uncertainty about foreign exchange rates (low-yield currency and high-yield currency separately). The third covariance term represents the risk premium arising from the uncertainty about the yield of the investment currency (there is no uncertainty about the funding yield, because the bond issued for funding expires at the end of the holding period). This expression clearly shows that, barring fortuitous cancellations, there is no expected excess return in the absence of risk premia. If the expected excess return is non zero, Eq 10 then allows us to attribute its value to different risk compensations. This requires, of course, the evaluation of the terms on the RHS of Eq 10. We show in what follows how these can be estimated.

If we define α , β , γ as the three covariance terms and a , b , c as the coefficients of these covariance terms, we can write Eq 10 more compactly as:

$$\mathbb{E}[xret] = a \cdot \alpha - b \cdot \beta - c \cdot \gamma + \Delta \quad (11)$$

We can simplify this expression further as follows. First, write for the future/forward FX ratio

$$\frac{FX_1(X, Y)}{FFX_0(X, Y; 1)} \equiv \Delta FX(X, Y) \quad (12)$$

Therefore we can write

$$\text{COV} \left[\frac{FX_1(X, Y)}{FFX_0(X, Y; 1)}, M_0^1(X) \right] = \text{COV} [\Delta FX(X, Y), M_0^1(X)] \quad (13)$$

Similarly we can define for the future/forward bond ratio

$$\Delta FPH_0^n \equiv \frac{PH_1^{n-1}}{FPH_0^n(1)} \quad (14)$$

which gives

$$\text{COV} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)}, M_0^1(X) \right] = \text{COV} [\Delta FPH_0^n, M_0^1(X)] \quad (15)$$

To gain financial and economic insight, let's assume next a simple CRRA utility function of the type

$$\begin{aligned} U(C) &= \frac{C^{1-RA} - 1}{1 - RA} \text{ for } RA \neq 1 \\ U(C) &= \log(C) \text{ for } RA = 1 \end{aligned} \quad (16)$$

(In the following to keep notation light we consider the logarithmic case. The same qualitative conclusions follow, and the equations we obtain simply differ by having consumption growth raised to the power RA .) With this utility function, the expression for the one-period stochastic discount factor is

$$M_0^1(X) = \delta \cdot \frac{U'(CX_1)}{U'(CX_0)} = \delta \cdot \frac{CX_0}{CX_1} = PX_0^1 \cdot g_{CX}^{-1} \quad (17)$$

where g_{CX} is consumption growth in currency X and δ is the one-period discount rate.

Putting the pieces together we have

$$\begin{aligned}
\mathbb{E}[xret(X, L, H, t; n)] = & \\
& -\text{COV}[\Delta FX(X, L), g_{CX}] \\
& +\mathbb{E}[\Delta FPH_0^n] \cdot \text{COV}[\Delta FX(X, H), g_{CX}] \\
& +\mathbb{E}[\Delta FX(X, H)] \cdot \text{COV}[\Delta FPH_0^n, g_{CX}]
\end{aligned} \tag{18}$$

This naturally leads us to an economic interpretation of the various terms that contribute to the profitability of the generalized carry strategy. If we assume rational asset pricing, from the perspective of an X investor the return from the generalized carry strategy can only differ from the risk-less return (*a priori* positively or negatively) if its payoffs (proportional to the the forwards/futures ratios) have non-zero covariance with the consumption growth in the X currency. The payoffs in turn can have a non-zero covariance with the X consumption growth if there is a *systematic* deviation of the forwards/futures ratio from 1. Next, we note that the ratios ΔFPH_0^n and $\Delta FX(X, H)$ are always positive, and therefore the profitability of the carry strategy depends on the signs of the covariance (risk premium) terms. Let's examine each term in turn. The expected return will be positive when

1. $\text{COV}[\Delta FX(X, L), g_{CX}] < 0$, ie, when the X currency strengthens with respect to the funding currency when the X economy strengthens.
2. $\text{COV}[\Delta FX(X, H), g_{CX}] > 0$, ie, when the X currency weakens with respect to the investment currency when the X economy strengthens.
3. $\text{COV}[\Delta FPH_0^n, g_{CX}] > 0$, ie, when the yields in the investment currency fall when the X economy strengthens.

These, of course, are just the conditions for the excess return to be positive. Whether the conditions are indeed satisfied is an empirical question. In practice, this means that, to answer the question of whether the generalized carry strategy can be expected, on average, to return more or less than the riskless rate, we must establish empirically whether there is a positive or negative correlation between the various forwards/futures ratios and the

consumption growth of the investor's currency. In order to answer this question, we estimate the various terms in Eq 18 by proceeding as follows.

4.4.1 Empirical Estimation of the α , β and γ Terms

As we have seen, the profitability of the generalized carry strategy that is the focus of this study depends on the terms

1. $\alpha = \text{COV} [\Delta FX_1(X, L), M_0^1(X)]$;
2. $\beta = \text{COV} [\Delta FX_1(X, H), M_0^1(X)]$; and
3. $\gamma = \text{COV} [\Delta FPH_0^n(1), M_0^1(X)]$

To measure the three covariance terms, we can consider the following two special strategies.

Special Strategy 1. First, consider the special carry strategy where $X = L \neq H$ and $n = 1$. $X = L$ implies $FX_1(X, L) = FF X_0(X, L; 1) = 1$. And $n = 1$ implies $PH_1^{n-1} = FPH_0^n(1) = 1$. Substituting into Eq 10, we obtain:

$$\mathbb{E} \left[\frac{FX_1(X, H)}{FF X_0(X, H; 1)} \right] = \frac{1}{PX_0^1} \text{COV} \left[\frac{FX_1(X, H)}{FF X_0(X, H; 1)}, M_0^1(X) \right] \quad (19)$$

From Eq 19, we can directly measure the β term, ie, covariance between the future-to-forward ratio of FX rate for H currency and the SDF. In our study, 12 such equations should be estimated separately, because there are 12 different X - H pairs (3 choices of H for each of the 4 investment currencies). Because H and L are just labels that can represent any currency among USD, EUR, GBP, and JPN, the 12 X - H pairs are the just same as the 12 X - L pairs. Therefore, we do not need to measure the α term in Eq 10 again, and $\alpha = \beta$. By calculating from our sample the average of the quantity $\left[\frac{FX_1(X, H)}{FF X_0(X, H; 1)} \right]$ (see below) we can obtain an unconditional (whole sample) estimate for the α covariance term on the RHS. We show below that a conditional estimate of α is also possible, but we defer the discussion for a later section.

Special Strategy 2. To measure the γ term, let's consider the second special strategy where $X = L \neq H$ and $n > 1$. For this strategy, we only have $FX_1(X, L) = FFX_0(X, L; 1) = 1$ implied by $X = L$. Substituting these conditions into Eq 18, we have:

$$\begin{aligned} & \mathbb{E} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)} \cdot \frac{FX_1(X, H)}{FFX_0(X, H; 1)} \right] - 1 = \\ & - \underbrace{\mathbb{E} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)} \right]}_b \cdot \underbrace{\frac{1}{PX_0^1} \text{COV} \left[\frac{FX_1(X, H)}{FFX_0(X, H; 1)}, M_0^1(X) \right]}_\beta \\ & - \underbrace{\mathbb{E} \left[\frac{FX_1(X, H)}{FFX_0(X, H; 1)} \right]}_c \cdot \underbrace{\frac{1}{PX_0^1} \text{COV} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)}, M_0^1(X) \right]}_\gamma \end{aligned} \quad (20)$$

From Eq 20 we see that we can obtain an unconditional estimate of the γ term by estimating the expectations using the sample averages of the respective ratios, $\Delta FX(X, H)$ and $\Delta FX(X, H)$, and by making use of the β terms that we have already calculated from Eq 19.

We present the empirical conclusions that can be drawn from Special Strategies 1 and 2 in Section 5.1. Before that, however, we extend the analysis by looking at the carry strategy from a conditional (state-dependent) perspective. This we do in the following Section 4.5.

4.5 From Unconditional to Conditional Estimates

So far, we have presented expressions for unconditional estimates of the α , β and γ terms. Greater insight can be gained by moving to a conditional estimate. The problem here is that on the LHS of Eq 18 we have the expectation of the excess return. Can we obtain an estimate for this quantity at each point in time (we call this a state-dependent estimate of the expectation)?

To make progress, we first simplify the analysis by estimating for each currency triplet the maturity-averaged return. This allows us to deal with $4 \times 4 \times 4 = 64$ strategies rather

than with $4 \times 4 \times 4 \times (n = 8) = 512$ strategies.⁷ Next, we note that, for all strategies, the realized excess returns appears to display a clear and strong mean-reverting behaviour. See Fig 3, which displays a few representative realizations of the unconditional excess returns process. We use an Augmented Dick-Fuller Test to test for unit roots, and we reject the unit root hypothesis for all the 64 (maturity-averaged) strategies, with a p value always smaller than 0.01.

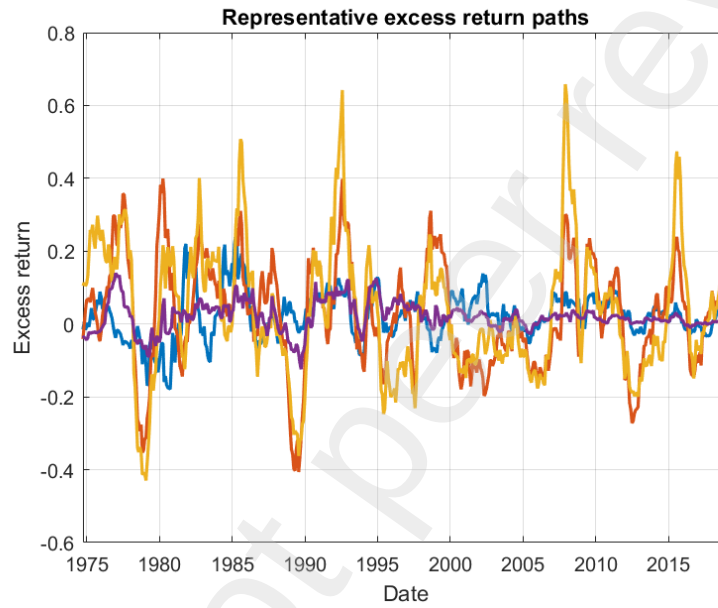


Figure 3: A few representative realizations of the unconditional excess returns process.

This prompts us to model the excess return process as an Ornstein-Uhlenbeck process,⁸ of the form

$$d(xret_t) = \kappa \cdot (\theta - xret_t) \cdot dt + \sigma \cdot dz_t \quad (21)$$

⁷The excess return for a triplet $\{X, L, H\}$ closely scales with the ‘duration’ n of the strategy, and therefore little extra information is added by considering the various maturities separately.

⁸We have explored the possibility that an auto-regressive model of order higher than 1 may provide better predictions. While for some strategies an AR(2) process provided a better fit to the data, out of sample better results were obtained using the simpler AR(1) model.

where $xret_t$ is the maturity-averaged excess return, κ is the mean-reversion speed, θ the mean-reversion level, σ the volatility of the process and dz_t the increment of a Brownian motion. (To keep notation light, we have denoted the maturity-averaged excess return simply by $xret_t$, but it should be $xret_t(X, L, H)$. Also the dependence on $\{X, L, H\}$ of κ and θ is suppressed, but is always implied.)⁹ When we discretize the process in Eq 21, and estimate the parameters of the associated AR(1) process, we find a very high R^2 (average value 0.89), and a highly significant regression coefficient (average t statistic of 67.13). We can therefore conclude that the process in Eq 21 gives a good representation of the behaviour of each excess return series. If we accept this process, the expectation of the excess return for a given strategy τ years later is given by

$$\mathbb{E}_t[xret(t + \tau)] = xret(t) \cdot \exp(-\kappa \cdot \tau) + \theta \cdot (1 - \exp(-\kappa \cdot \tau)) \quad (22)$$

where we have emphasized with the subscript in $\mathbb{E}_t[\cdot]$ that this is now a time-dependent (conditional) expectation. We will use this expression later on to estimate a return-predicting factor for the strategy (see Section 5.4). For the moment, we simply use this result to estimate, for each strategy, the expected return over the holding period of the strategy (12 months). This gives us the conditional expectation needed to produce a conditional estimate of all the covariance terms in Eq 18. In doing so, we assume that investors observe, as we do, the last realized excess return, realize that the excess return process is mean-reverting, estimate (out of sample) its parameters, and derive the expectation of the next period excess return using Eq 22.

This is important, because, if there is some predictability in the excess return process (and mean reversion is one such instance of predictability), then it should be possible to exploit this predictability to establish a state-dependent strategy. To do so, the investor can start from the unconditional strategy, in which she always invests the same amount, irrespective of market conditions. However, if the investor really believes the excess return

⁹In what follows, unless there is the possibility of ambiguity, we call the maturity-averaged excess return simply the excess return.

process to be mean-reverting, she can scale the size of her strategy by the next-period expected excess return. More precisely, at time t the investor observes the excess return of the carry strategy that has just come to maturity, calculates from this realized value the expected return for the same strategy over the next period, and sets the notional for the same strategy over the next holding period equal to this expected excess return. The procedure, which is fully no-peek-ahead, is repeated for each of the 64 strategies. We present the results in Section 5.4, where we compare them with the unconditional results.

5 Results

We present and discuss separately the result for the unconditional and the conditional strategies in Sections 5.1 and 5.4, respectively.

5.1 Unconditional Strategy – Sharpe Ratio and Returns

We recall that with the unconditional strategy the same initial notional is put in place irrespective of the state. We have considered both the case when the holding period is monthly and when it is yearly. Since we obtain similar results, in this section we only report the case of yearly results.¹⁰ Figs 4 and 5 show the excess returns and Sharpe ratios for the nine maturities and the 64 strategies. The last column shows the average across maturities of the excess return or Sharpe Ratio, as appropriate. The first column provides the currency of the investor (X), the funding currency (L) and the investment currency (H). The entries have been sorted in descending order from highest to lowest average excess return or Sharpe ratio, as appropriate.

These tables raise the key question of whether the flexibility offered by changing the funding and investment currency offers a higher excess return or Sharpe Ratio. To answer this question quantitatively we build a ‘diversity index’, $div(X, L, H)$, which we set equal

¹⁰The average and standard deviation of the 64 Sharpe Ratios for the unconditional strategy were 0.226 and 0.124, respectively for the 12-month holding period, and 0.262 and 0.137 for monthly holding period.

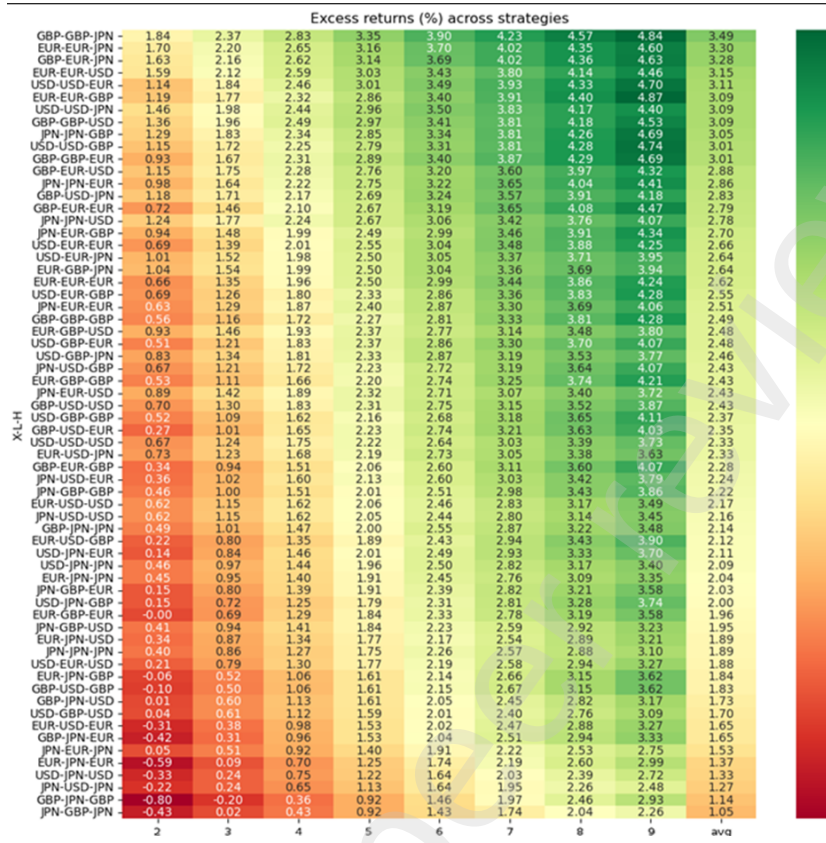


Figure 4: The excess returns for the nine maturities and the 64 strategies. The last column shows the average across maturities of the excess return. The first column provides the currency of the investor (X), the funding currency (L) and the investment currency (H). The entries have been sorted in descending order from highest to lowest average excess return.

to 0 if the currency of the investor, of funding and of investing are all equal; to 1 if one currency is different; and to 2 if all currencies are different. To put this indicator in context, we recall that the traditional interest-rate carry strategy has a diversity index of 0, and that the UIP carry strategy has an index of 1. Having built this indicator, we regress the maturity-averaged Sharpe Ratios, $\langle SR(X, L, H) \rangle$, against the diversity index:

$$\langle SR(X, L, H) \rangle = a + b \cdot div(X, L, H) + \epsilon \quad (23)$$

When we do so we obtain a highly significant b coefficient (t statistic = -4.41), an R^2 of 0.25, and a negative dependence of the average Sharpe ratio on the diversity index. See

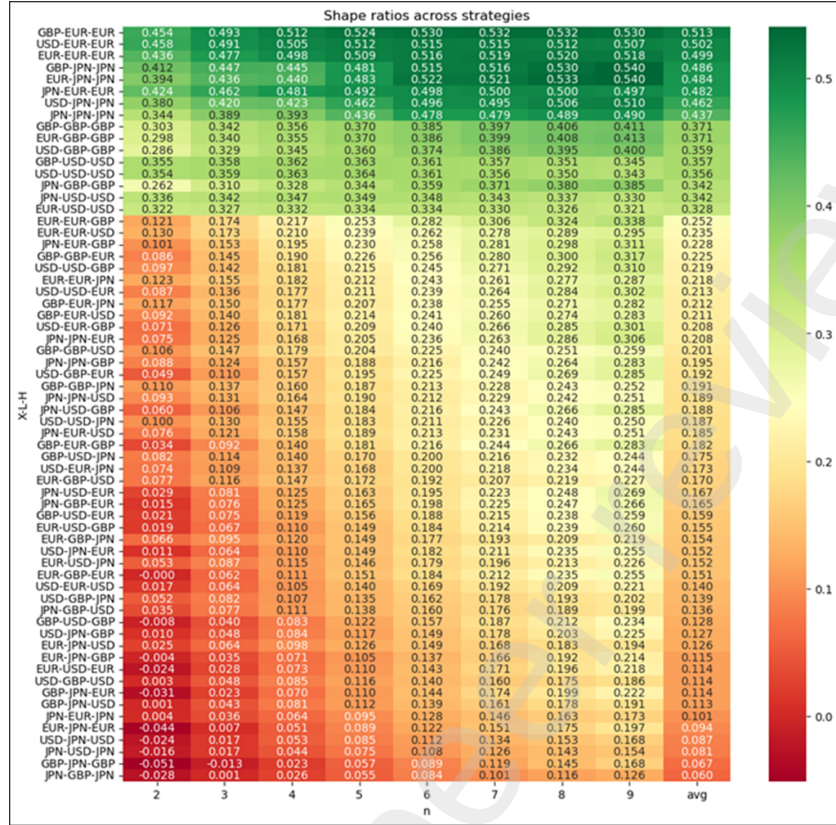


Figure 5: The Sharpe Ratio for the nine maturities and the 64 strategies. The last column shows the average across maturities of the excess return. The first column provides the currency of the investor (X), the funding currency (L) and the investment currency (H). The entries have been sorted in descending order from highest to lowest average Sharpe ratio.

the scatter plot in Fig 6, which shows the average Sharpe Ratio plotted as a function of the diversity index. For future reference, we note that this figure should be compared and contrasted with Fig 13, which refers to the conditional strategy, and for which, as we will explain, the b coefficient is positive.

When we run a similar regression, but using as left-hand variable instead of the Sharpe Ratios the maturity-averaged excess returns, $\langle xret(X, L < H) \rangle$,

$$\langle xret(X, L, H) \rangle = A + B \cdot div(X, L, H) + \eta \quad (24)$$

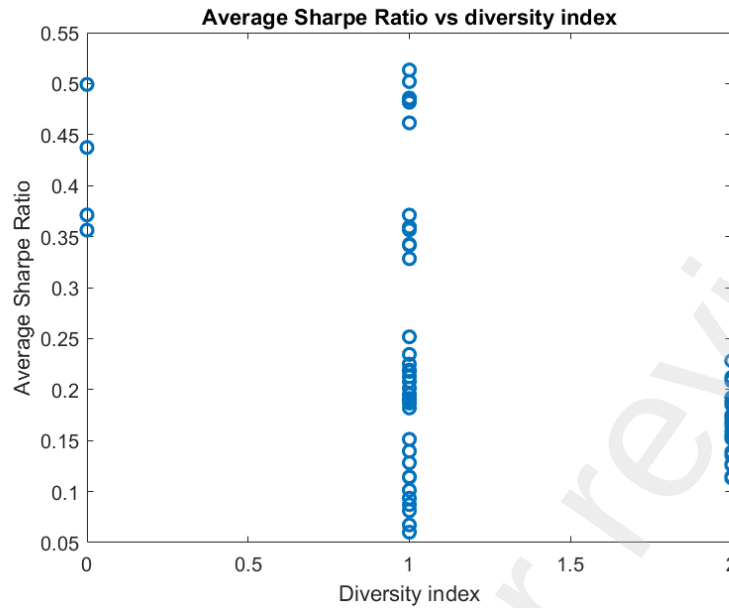


Figure 6: The maturity-averaged Sharpe Ratio plotted as a function of the diversity index.

the R^2 is virtually zero ($3 \cdot 10^{-6}$) and the t statistic for the B coefficient of 0.005, showing that the diversity index now does not explain excess returns. We can therefore conclude that diversifying the investment and funding currencies *per se* does not add to expected excess returns, and is actually detrimental to the Sharpe Ratio. This happens because the added volatility that arises when investor X funds and or invests in currencies other than X tends to negate the potential advantage of extracting a higher return. This is *a priori* surprising: it is well known, for instance (and borne out by our analysis), that the traditional single-currency interest-rate carry strategy is very profitable, and has a high Sharpe Ratio, for USD. It would therefore seem natural for a non-USD investor to exploit this higher Sharpe Ratio by exchanging her currency into USD and funding and investing in USD as the USD-denominated investor does. What we find is that her quest for a high Sharpe Ratio would be frustrated because of the extra volatility introduced by the need to exchange the dollar net profits in her own currency. We will revisit these conclusions when we discuss the results obtained with the conditional strategy presented in Section 5.4.

This observation also puts in perspective the popular carry strategy of funding oneself in the currency with the lowest yields and investing in the currency offering the highest yields. In terms of Sharpe Ratios, the strategy may work if the domestic currency coincides with either the funding or the investment currency, but is unlikely to be attractive if the three currencies are all different (ie, for the case $X \neq L \neq H$). We will quantify this intuition in Section 5.4.

The next observation is that, at least once averaged over maturities, all the 64 unconditional strategies have positive excess returns. This prompts the questions: Where does this profitability come from? What are investors compensated for? We answer this question in the following section.

5.2 Analysis of Profitability

Recall that the expected profitability of the generalized carry strategy is given by Eq 10, reproduced below for ease of reference:

$$\begin{aligned}
& -\mathbb{E} \left[\frac{FX_1(X, L)}{FFX_0(X, L; 1)} \right] + \mathbb{E} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)} \cdot \frac{FX_1(X, H)}{FFX_0(X, H; 1)} \right] = \\
& \underbrace{\frac{1}{PX_0^1}}_a \underbrace{\text{COV} \left[\frac{FX_1(X, L)}{FFX_0(X, L; 1)}, M_0^1(X) \right]}_\alpha \\
& - \underbrace{\mathbb{E} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)} \right]}_b \cdot \underbrace{\frac{1}{PX_0^1} \text{COV} \left[\frac{FX_1(X, H)}{FFX_0(X, H; 1)}, M_0^1(X) \right]}_\beta \\
& - \underbrace{\mathbb{E} \left[\frac{FX_1(X, H)}{FFX_0(X, H; 1)} \right]}_c \cdot \underbrace{\frac{1}{PX_0^1} \text{COV} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)}, M_0^1(X) \right]}_\gamma + \Delta
\end{aligned} \tag{25}$$

In order to understand the relative importance of the different contributors to profitability, we analyze separately the various components, a , α , b , β , c and γ .

	2	3	4	5	6	7	8	9
USD	1.0063	1.0117	1.0166	1.0211	1.0252	1.0289	1.0323	1.0355
GBP	1.0051	1.0107	1.0160	1.0211	1.0262	1.0310	1.0355	1.0399
EUR	1.0063	1.0128	1.0187	1.0239	1.0287	1.0331	1.0371	1.0408
JPN	1.0038	1.0082	1.0123	1.0169	1.0219	1.0249	1.0280	1.0301

Table 4: The ratio $\frac{PH_1^{n-1}}{FPH_0^n(1)}$, ie, the quantity b in Eq 25, times PX_0^1 , for initial maturities 2 to 9 years. 12-month holding period.

The term a is straightforward and relatively unimportant. Therefore we do not comment on this term any further. We calculate the unconditional b and c terms by time averaging the respective quantities over the whole sample. The results are presented in Tabs 4 and 5. (In Tab 4 we present $b \cdot PX_0^1$, as this quantity immediately shows whether the interest-rate EH is satisfied or not, ie, whether forward bond prices are unbiased expectations of future bond prices.) The first observation is that both the b and c quantities are very close to 1, and therefore are not the drivers of the profitability of the generalized strategies we analyze in this study.

Since, as we have shown, the a , b and c terms are not the drivers of the profitability of the generalized carry strategy, we move to the analysis of the terms α , β and γ . We note that the α and β terms have opposite-sign contributions to the profitability (as one is associated with the funding and one with the investment leg), but the covariances are the same irrespective of whether they appear in the funding or investment term. We therefore report in Tab 6 only the quantity β . In order to estimate this β term, we follow Special Strategy 1, by time-averaging the realized returns for this special strategy, and using the value b reported above. The results are shown in Tab 6. We note that for all currency pairs the β (α) term is negative. Therefore it will provide a positive contribution to profitability for the investment leg, and a negative contribution for the funding leg. Since, as we discuss at greater length below, the duration of the investment leg is always greater or equal to

	USD	GBP	EUR	JPN
USD	1.053	1.088	1.042	1.014
GBP	1.045	1.067	1.025	1.003
EUR	1.068	1.094	1.043	1.022
JPN	1.085	1.118	1.067	1.031
USD	1.000	1.019	0.999	0.985
GBP	0.993	1.000	0.984	0.975
EUR	1.014	1.025	1.000	0.993
JPN	1.030	1.047	1.023	1.000

Table 5: The term c in Eq 25 (first four rows), and the ratio $\frac{FX_1(X,H)}{FFX(X,H;1)}$ (next four rows), time-averaged over the whole sample. 12-month holding period.

the duration of the funding leg, the net contribution of the negative β terms is positive: investors are compensated for receiving negative (positive) payoff from the FX components of the strategy when their economy is doing badly (well).

Next we move to the term γ . To estimate this quantity we follow Special Strategy 2, and make use of the already-estimated $b \cdot \beta$ term – see Eq 20. Since the term c is always positive and independent of the maturity, n , the sign and relative magnitude across maturities of the term $c \cdot \gamma$ then depends on the covariance term $\gamma(X, H; n) = \text{COV} \left[\frac{PH_1^{n-1}}{FPH_0^n(1)}, M_0^1(X) \right]$. By averaging expected returns over the whole sample, we obtain an unconditional estimate of the sign and relative magnitude of the term $\gamma(X, H; n)$. This is shown in Tab 7: for each investment currency, X , there are four such terms, one of which corresponds to the case $X = H$ (which is the traditional interest-rate carry strategy). Fig 7 displays the same information in graphical form by showing the unconditional (whole sample) estimate of the term $\gamma(X, H; n)$ for the four possible domestic currencies (X), averaged over the four investment currencies, as a function of the maturity of the investment bond.

H(L)	X	β (α)
EUR	EUR	0
EUR	GBP	-0.002
EUR	JPN	-0.005
EUR	USD	-0.005
GBP	EUR	-0.007
GBP	GBP	0
GBP	JPN	-0.010
GBP	USD	-0.006
JPN	EUR	-0.011
JPN	GBP	-0.011
JPN	JPN	0
JPN	USD	-0.009
USD	EUR	-0.009
USD	GBP	-0.006
USD	JPN	-0.008
USD	USD	0

Table 6: The term β in Eq 18, time-averaged over the whole sample, for the domestic currency, X , and the investment or funding currency, H or L . The same estimate hold for 12-month holding period.

We are now in a position to gain a full appreciation of the sources of profitability of the generalized carry strategies: broadly speaking, the profitability is driven, in descending order of importance, by the $c \cdot \gamma$, followed by the $b \cdot \beta$ term and then, as a distant third, by the $a \cdot \alpha$ term. We also note that, for all strategies, excess returns are a monotonically increasing function of maturity, n . See, in this respect, Fig 7, which shows the unconditional (whole sample) estimate of the term $\gamma(X, H; n)$ for the four possible domestic currencies (X), averaged over the four investment currencies, as a function of the maturity of the investment bond.

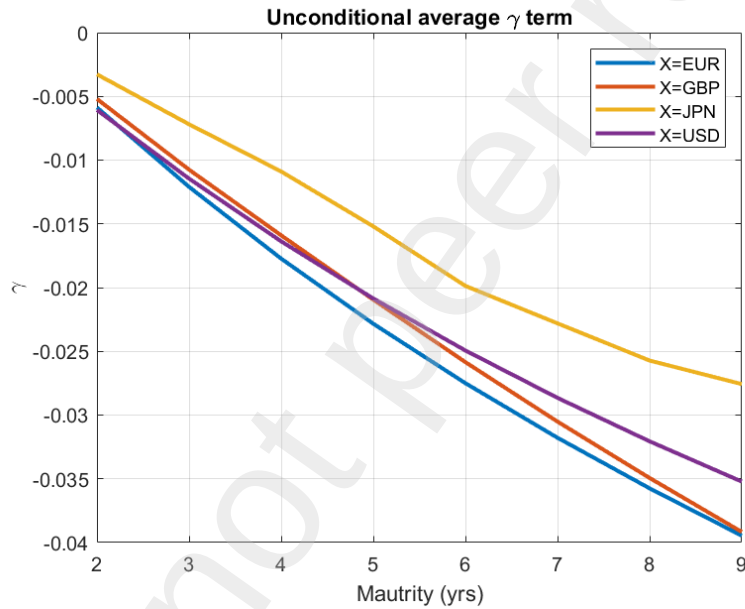


Figure 7: The unconditional (whole sample) estimate of the term $\gamma(X, H; n)$ for the four possible domestic currencies (X), averaged over the four investment currencies, as a function of the maturity of the investment bond.

This suggests that the bulk of the profitability mainly comes from the interest-rate component of the mixed strategies. We also note from Tab 5 that the contribution to the profitability from the c term is small. Therefore the most important contribution in the term $-c \cdot \gamma$ comes from γ . This generalizes to the multi-currency setting what has been

found in the setting of a same-currency interest-rate carry strategies, ie, that the expected return is always on average positive and that there is, on average, a positive ‘duration’ risk premium (see, eg, the discussion in Rebonato (2018)¹¹). Recall from Section 4 that a negative γ terms (which contributes positively to the profitability of the generalized carry strategy) would correspond to a compensation for bearing consumption risk when interest rates are falling. If, as it is customary to do in empirical asset pricing studies, one takes the return from a diversified equity index as a proxy for consumption growth (see, eg, Seiler (2005) for a recent discussion), we see that a negative γ term corresponds to the case of a *positive* correlation between Treasury and equity returns. With the exception of the period from the late 1990s to the mid-2010s, this has empirically been the case (see the discussion in Adrain, Kramer, and Malik (2026) in this respect), and this observation has been adduced as an explanation for the on-average-positive bond risk premium in the same-currency carry strategies in USD, GBP, EUR and JPN. What we establish in this study generalizes this finding to a multi-currency setting. The strong correlation between yields in different currencies shown in Section 3 then explains why this generalization holds.

Next we note that, while a negative γ term always contributes to profitability, the contribution from the β (α) term requires a more careful discussion. Empirically, Tab 6 shows that the term is negative. Whether this negative term contributes positively or negatively to profitability depends whether it appears in the funding or investment leg. The two terms, however, are not symmetric: the covariance between the domestic discount factor and the FX forward/futures ratio is larger in the investment leg (whose maturity ranges from 1 to 9 years) than in the funding leg (whose maturity is always 1 year). Therefore, overall the negative β (α) term has a positive contribution to profitability.

¹¹See, in particular, Chapters 23 to 25.

H	X	2	3	4	5	6	7	8	9
EUR	EUR	-0.0060	-0.0122	-0.0178	-0.0229	-0.0275	-0.0318	-0.0357	-0.0393
EUR	GBP	-0.0055	-0.0114	-0.0168	-0.0217	-0.0262	-0.0304	-0.0342	-0.0378
EUR	JPN	-0.0063	-0.0128	-0.0187	-0.0240	-0.0288	-0.0332	-0.0373	-0.0411
EUR	USD	-0.0058	-0.0120	-0.0176	-0.0227	-0.0274	-0.0318	-0.0359	-0.0397
GBP	EUR	-0.0051	-0.0105	-0.0156	-0.0205	-0.0253	-0.0298	-0.0341	-0.0382
GBP	GBP	-0.0048	-0.0100	-0.0149	-0.0197	-0.0244	-0.0289	-0.0332	-0.0373
GBP	JPN	-0.0058	-0.0117	-0.0172	-0.0225	-0.0277	-0.0327	-0.0373	-0.0418
GBP	USD	-0.0052	-0.0108	-0.0160	-0.0210	-0.0260	-0.0306	-0.0351	-0.0393
JPN	EUR	-0.0032	-0.0071	-0.0108	-0.0150	-0.0196	-0.0226	-0.0255	-0.0272
JPN	GBP	-0.0030	-0.0067	-0.0103	-0.0145	-0.0191	-0.0220	-0.0248	-0.0264
JPN	JPN	-0.0037	-0.0079	-0.0119	-0.0164	-0.0212	-0.0242	-0.0272	-0.0293
JPN	USD	-0.0032	-0.0071	-0.0107	-0.0149	-0.0196	-0.0225	-0.0254	-0.0273
USD	EUR	-0.0064	-0.0120	-0.0171	-0.0218	-0.0260	-0.0298	-0.0333	-0.0366
USD	GBP	-0.0057	-0.0107	-0.0153	-0.0195	-0.0234	-0.0269	-0.0301	-0.0330
USD	JPN	-0.0064	-0.0121	-0.0173	-0.0220	-0.0264	-0.0303	-0.0340	-0.0373
USD	USD	-0.0059	-0.0111	-0.0158	-0.0201	-0.0240	-0.0276	-0.0309	-0.0339

Table 7: The unconditional (whole sample) estimate of the term $\gamma(X, H; n)$ for the four possible domestic currencies (X), and the four possible investment currencies H , as a function of the maturity (in years, top row) of the investment bond.

5.3 Generalized and Traditional Carry Strategies Compared

One could ask whether the generalized carry strategy we have analyzed differs significantly from the sum of an interest-rate-only and an FX-only carry strategy. The difference would be important if there were a significant covariance between the returns of the two strategies. To answer this question, we look at the ϵ^n term, which is defined as the difference between the excess return of our strategy with investment maturity n and the sum of two traditional carry strategies, which, with obvious notation, we call $xret_{FX}^1$ and $xret_{ir}^n$:

$$\epsilon^n = xret^n - (xret_{FX}^1 + xret_{ir}^n) \quad (26)$$

where n is the maturity of the investment bonds. If the excess returns of our strategy were equal to the sum of the two traditional strategies, then the ϵ^n term should not be significantly different from zero. We present below our estimates of the quantity ϵ^n for the strategies that invest in the 9-year maturity bonds, ie, $n = 9$. The excess return of our carry strategy and the two traditional carries, as well as the ϵ terms, are presented in Table 8.

H	X	$xret$	$xret_{ir}$	$xret_{FX}$	Sum	ϵ
EUR	USD	0.0460	0.0360	0.0050	0.0410	0.0060
GBP	USD	0.0470	0.0360	0.0070	0.0430	0.0050
JPN	USD	0.0440	0.0360	0.0100	0.0460	-0.0020
USD	EUR	0.0440	0.0410	0.0090	0.0500	-0.0060
GBP	EUR	0.0480	0.0410	0.0070	0.0480	0.0010
JPN	EUR	0.0450	0.0410	0.0120	0.0530	-0.0070
USD	GBP	0.0440	0.0400	0.0060	0.0460	-0.0020
EUR	GBP	0.0460	0.0400	0.0020	0.0420	0.0040
JPN	GBP	0.0470	0.0400	0.0120	0.0520	-0.0050
USD	JPN	0.0410	0.0300	0.0070	0.0370	0.0040
EUR	JPN	0.0440	0.0300	0.0050	0.0350	0.0090
GBP	JPN	0.0470	0.0300	0.0100	0.0400	0.0070

Table 8: The excess returns of the generalized carry strategy and of the two traditional carry strategies, as well as the excess return difference between our strategy and the sum of the two traditional carries. Each row presents the results for one H-X currency pair strategy. The H column presents the investment currency. The column presents the domestic currency. The column $xret$ summarizes the excess return/risk premia of our strategy with 9-y maturity of investment bonds. The column $xret_{ir}$ shows the excess return of the traditional interest-rate carry (funding 1y and investing 9-y) for investors in currency X. The column $xret_{FX}$ presents the excess return of the traditional FX carry (funding 1-y in currency X and investing 1-y in currency H). The column labelled 'Sum' is the sum of the two traditional carries. The last column ϵ is the difference between the excess return difference between our strategy and the sum of the two traditional carries, as calculated in Eq 26.

Tab 8 presents the estimate of the quantity ϵ from the historical data, but does not

allow us to make any statistical inference about its significance. In order to conduct formal statistical-significance tests, we employ the non-parametric resampling procedure in Crump and Gospodinov (2019). We choose this approach because i) it is specifically designed to investigate the factor structure of interest rates yield curves; ii) it addresses the challenges in characterizing the minimal dimension of the data-generating process; iii) it is robust to general forms of time and cross-sectional dependence as well as conditional heteroscedasticity of unknown form; and iv) it is non-parametric, and therefore does not rely for its validity on the correctness of a specific data-generating model. This resampling technique and the tests of its performance for the data at hand are discussed in detail in Appendix A.

For the problem at hand we use the Crump and Gospodinov (2019) technique by conducting 1,000 resamplings and creating a distribution for the ϵ term. We can see that the ϵ term is significant at 90% confidence level for all but two of the 12 H-X currency pairs, and at the 95% confidence level for more than half of the pairs. Tabs 9 and 10 present the resampled results of the decomposed terms for the strategy with the longest investment bond maturity (9 years).

H	X	$xret$	$-b\beta$	$-c\theta$	$-b\beta - c\theta$	Δ	$xret_{ir}$	sum	ϵ
EUR	USD	0.033	0.002	0.031	0.033	0.001	0.030	0.032	0.001
		(0.02)	(0.001)	(0.02)	(0.02)	(0.001)	(0.02)	(0.02)	(0.001)
GBP	USD	0.037	0.003	0.033	0.036	0.001	0.030	0.033	0.004**
		(0.03)	(0.002)	(0.02)	(0.03)	(0.001)	(0.02)	(0.02)	(0.002)
JPN	USD	0.027	0.005	0.020	0.025	0.002	0.030	0.035	-0.008**
		(0.01)	(0.002)	(0.01)	(0.01)	(0.000)	(0.02)	(0.02)	(0.004)
USD	EUR	0.031	0.005	0.024	0.029	0.002	0.025	0.030	0.001
		(0.02)	(0.002)	(0.02)	(0.02)	(0.001)	(0.02)	(0.02)	(0.001)
GBP	EUR	0.035	0.003	0.031	0.034	0.001	0.025	0.028	-0.007**
		(0.02)	(0.002)	(0.02)	(0.02)	(0.000)	(0.02)	(0.02)	(0.004)
JPN	EUR	0.029	0.007	0.021	0.028	0.001	0.025	0.032	-0.003*
		(0.02)	(0.003)	(0.02)	(0.02)	(0.001)	(0.01)	(0.02)	(0.002)

Table 9: The Crump-Gospodinov-resampled decomposition of the excess return for the strategy with a 9-year bond maturity for all 6 of the currency pairs based on Eq.11, as well as a comparison of our strategy with the sum of two traditional carry trades. The results are based on 1000 resamplings. Each row presents the results for one H-D currency pair strategy. The H column presents the investment currency. The X column presents the domestic currency. The column “xret” presents the expected excess returns of our strategy. The column $xret_{ir}$ presents the expected excess returns (risk premia) of the traditional interest rate carry strategy. The $-b\cdot\beta$ term is just the expected excess returns (risk premia) of the traditional interest FX carry strategy. The “sum” column is the sum of two traditional carries. For the ϵ column, the triple asterisk (***), double asterisk (**) and asterisk (*) represent statistical significance at the 99%, 95%, and 90% levels, respectively.

a

H	X	$xret$	$-b\beta$	$-c\theta$	$-b\beta - c\theta$	Δ	$xret_{ir}$	sum	ϵ
USD	GBP	0.018	0.005	0.011	0.016	0.002	0.023	0.029	-0.009**
		(0.01)	(0.003)	(0.01)	(0.01)	(0.002)	(0.01)	(0.01)	(0.005)
EUR	GBP	0.021	0.004	0.016	0.020	0.001	0.023	0.027	-0.006**
		(0.01)	(0.002)	(0.01)	(0.01)	(0.001)	(0.01)	(0.01)	(0.003)
JPN	GBP	0.024	0.006	0.017	0.024	0.000	0.023	0.029	-0.005*
		(0.02)	(0.003)	(0.02)	(0.02)	(0.001)	(0.02)	(0.02)	(0.003)
USD	JPN	0.018	0.004	0.013	0.017	0.001	0.019	0.023	-0.005**
		(0.01)	(0.003)	(0.01)	(0.01)	(0.000)	(0.01)	(0.01)	(0.002)
EUR	JPN	0.027	0.004	0.021	0.025	0.002	0.019	0.023	0.004*
		(0.01)	(0.002)	(0.01)	(0.01)	(0.001)	(0.01)	(0.01)	(0.002)
GBP	JPN	0.032	0.005	0.025	0.030	0.002	0.019	0.024	0.008**
		(0.02)	(0.003)	(0.01)	(0.02)	(0.000)	(0.01)	(0.02)	(0.004)

Table 10: Same as Tab 9 for the remaining 6 currency pairs

So, we can conclude that, for the highly correlated currencies covered in our study, the difference between the generalized carry strategy and the sum of the two traditional strategies is often statistically significant. For several strategies it is also economically significant, as it can be as large as 50% of the total excess return (for the case $X = GBP$ and $H = USD$). See Fig 8, which shows the ratio of the difference term ϵ^n defined in Eq 26 to the total excess return, for $n = 9$ and for selected X/H combinations.

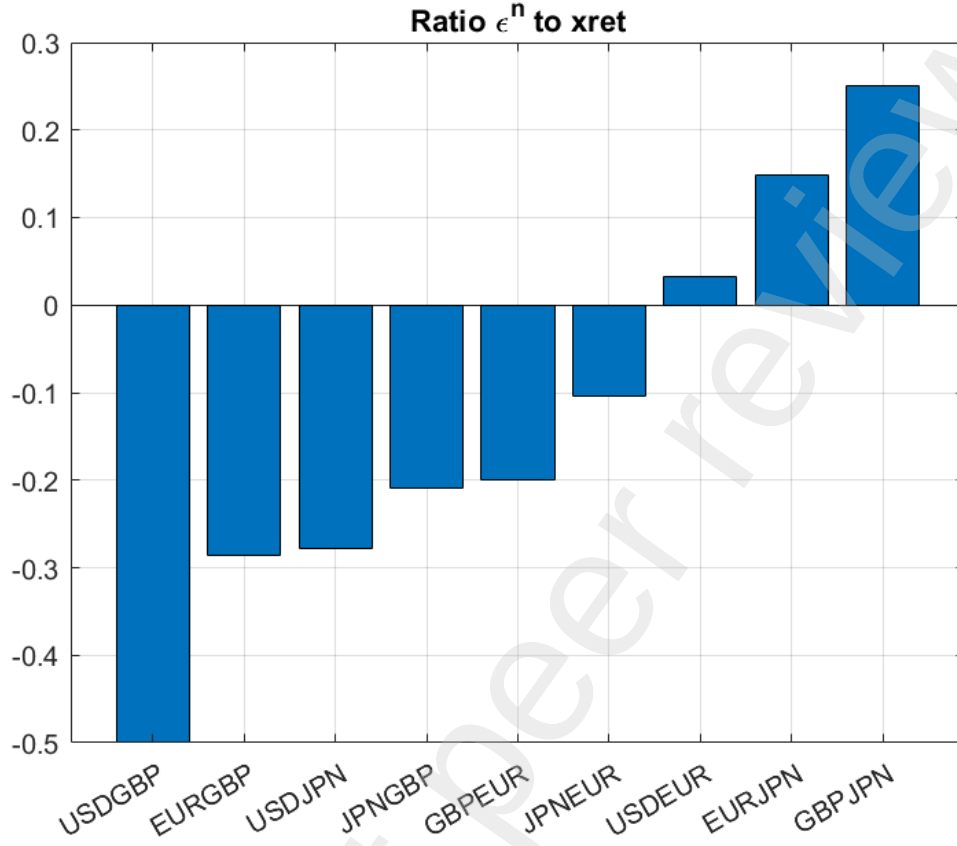


Figure 8: The ratio of the difference term ϵ^n defined in Eq 26 to the total excess return, for $n = 9$ and for selected X/H combinations.

We want to understand where the difference between the generalized strategy and the sum of the FX and interest-rate carry strategies comes from. To do this, we can decompose the expectation of the ϵ^n term into the difference of covariances. More precisely, taking an expectation of Eq 26, we have

$$\mathbb{E}[\epsilon^n] = \mathbb{E}[xret^n] - (\mathbb{E}[xret_{FX}^1] + \mathbb{E}[xret_{ir}^n]) \quad (27)$$

On the RHS, the first term, $xret^n$, is the expected excess return of our strategy ($X \neq H, n > 1$), which is the RHS of Eq 20. The second term, $xret_{FX}^1$, is just the excess return of the traditional UIP strategy ($X \neq H, n = 1$), whose expected excess return is shown on

the RHS of Eq 19. The last term, $xret_{ir}^n$, comes from the traditional interest rate carry strategy, whose excess return can be obtained by setting $H = L = X$ in Eq 10:

$$\mathbb{E}[xret_{ir}^n] = \mathbb{E}\left[\frac{PX_1^{n-1}}{FPX_0^n(1)}\right] = 1 - \frac{1}{PX_0^1} \text{COV}\left[\frac{PX_1^{n-1}}{FPX_0^n(1)}, M_0^1(X)\right] \quad (28)$$

Substituting the three expected excess returns into Eq 27, we have

$$\mathbb{E}[\epsilon^n] = \frac{1}{PX_0^1} \left(\text{COV}\left[\frac{PX_1^{n-1}}{FPX_0^n(1)}, M_0^1(X)\right] - \text{COV}\left[\frac{PH_1^{n-1}}{FPH_0^n(1)}, M_0^1(X)\right] \right) \quad (29)$$

Making use of Eq 17 and of the definitions of the forwards/future ratios, and still considering for notational simplicity the case of a coefficient of relative risk aversion of 1, Eq 29 can be re-written as

$$\mathbb{E}[\epsilon^n] = \text{COV}[\Delta FPH^n, g_{cX}] - \text{COV}[\Delta FPX^n, g_{cX}] \quad (30)$$

So, in economic terms, the quantity ϵ_n will be small i) if there little correlation between the growth in the domestic economy (g_{cX}) and both the yields in the investment currency (the term ΔFPH^n) and funding currency (the term ΔFPX^n), or ii) if the correlations (and the volatilities) are similar. Indeed, we note that the largest, second largest and fourth largest, and the second, third and fourth most negative values for the term ϵ^9 (see Fig 8) involve the currency JPN, whose yields, as shown in Section 3, display the lowest correlation with all the other currencies.

5.4 The Conditional Strategy

So far we have analyzed unconditional carry strategies, by which we mean that the investment in the H bond is always of the same sign and notional. We now move to the analysis of the conditional strategy, briefly introduced at the end of Section 4.5. The idea is that, if the investor can establish that there is predictability in the unconditional excess return process, then she can scale the notional of her unconditional carry strategy based on her estimate of the expected excess return over the next investment period. We stress that, if she predicts the excess return over the next holding period to be negative, she can reverse the sign of her investment and funding positions. In the simple model we present, her predictions are made assuming that the excess return process follows the Ornstein-Uhlenbeck

process in Eq 21, for which the expectation is given by Eq 22. The reversion-speed and reversion-level parameters of the process are estimated using past realizations of the excess return process. However, the analysis reported so far has looked at overlapping strategies put in place every month and held for 12 months. If the holding period were kept at 12 months, too much variance would be superimposed on the expectation 12 months hence for it to retain enough information by the end of the holding period. For the conditional analysis we therefore shorten the holding period to one month – which is, after all, a more realistic holding period for a professional investor.

As mentioned, as far as the return and Sharpe Ratio properties of the unconditional strategy are concerned, qualitatively very similar results hold in the case of yearly and monthly holding periods. Since the unconditional analysis presented so far remains in essence valid for the shorter holding period, we do not duplicate the presentation of the results. What do change dramatically are the return and Sharpe Ratio properties of the conditional strategy, and it is to the discussion of these statistics that we now turn.

We conducted two types of analysis: one descriptive in nature, in which the regressions to estimate the reversion speeds and reversion levels were carried using all-sample coefficients (this follows procedure to construct return-predicting factors in papers such as Cochrane and Piazzesi (2005) or Cieslak and Povala (2015)); and one predictive in nature, in which the same coefficients are estimated using an ever-expanding window, without any peak-ahead. We refer to the second set of estimates the ‘predictive regressions’.¹² To be clear: with the descriptive regression, at time T the investor has no information about future yields and returns, but knows the all-sample reversion coefficients. With the predic-

¹²We have used the first 24 months of data to carry out the first estimate of the regression coefficients, and then progressively expanded the window by one month every month. The excess return record for the predictive study is therefore shorter by two years. For consistency we also compare the unconditional Sharpe Ratios without the first two years’ worth of data. This is why the histograms for unconditional Sharpe Ratios look marginally different in the descriptive and predictive cases.

tive analysis the same investor still has no information about future yields, but also has to estimate the mean-reversion parameters using past data only.

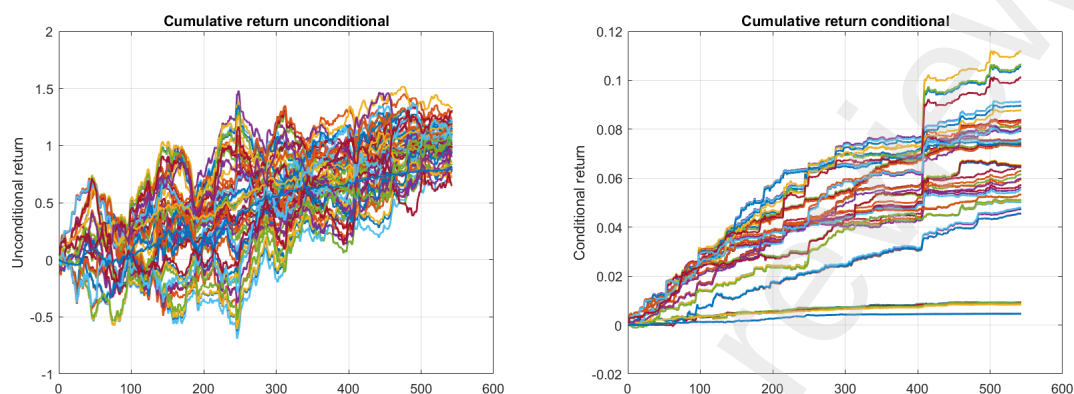


Figure 9: The cumulative returns for the 64 unconditional (left panel) and conditional (right panel) strategies, averaged across maturities. All-sample regression.

To begin with, Fig 9 shows the cumulative returns for the 64 unconditional (left panel) and conditional (right panel) strategies, averaged across maturities for the descriptive regression. Before embarking in any formal analysis, it is clear that the behaviour is strikingly different: in particular, since with the conditional carry strategy the investor can be long or short, she could, if her predictions were reliable enough, avoid periods of negative returns. The right panel of Fig 9 shows that her return predictions are indeed very informative, and correctly tell her when to switch the sign of the positions. As a result, we do not observe any prolonged periods when the cumulative returns decline. The same Fig 9 also suggests that drawdowns should be much smaller for the conditional than the unconditional strategies. Indeed, Fig 10, which shows the maximum drawdowns for the unconditional and the conditional strategies, confirms that this is indeed the case.¹³ This feature makes the

¹³To carry out a fair comparison, the notionals of each of the 64 conditional strategies were scaled so as to have the same final cumulative profitability as the corresponding unconditional strategy. Scaling by a constant does not change the Sharpe Ratio, discussed below.

conditional strategy of great interest to institutional and regulated investors, who can face drawdown, stop-loss and Value at Risk limits or similar constraints.

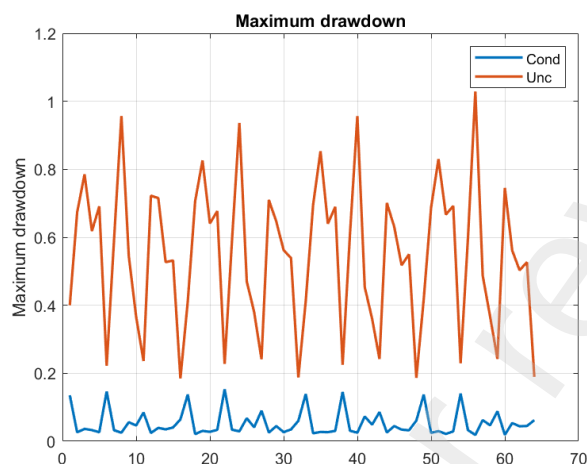


Figure 10: The maximum drawdowns for the unconditional and the conditional strategies (labelled 'Unc' and 'Cond', respectively). The notionals of each of the 64 conditional strategies were scaled so as to have the same final cumulative profitability as the corresponding unconditional strategy.

This marked difference in performance is reflected in the Sharpe Ratios of the two strategies, shown in Figs 11 and Figs 12, which display the histograms of unconditional and conditional Sharpe Ratios for the 64 average carry strategies discussed in the text (left panel) and the difference between the two (conditional - unconditional) (right panel) for the descriptive and predictive regressions, respectively. With either strategy, for the vast majority of the strategies (for 81% and for 75% of the strategies for the descriptive and predictive analysis, respectively), switching from the unconditional to the conditional strategy increases the Sharpe Ratio, and the average increase, $\langle \Delta SR \rangle$, is highly statistically and economically significant: $\langle \Delta SR \rangle = 0.46$ and $\langle \Delta SR \rangle = 0.41$, again for the descriptive and predictive analysis, respectively.

It is also very interesting to regress the average conditional excess returns and Sharpe

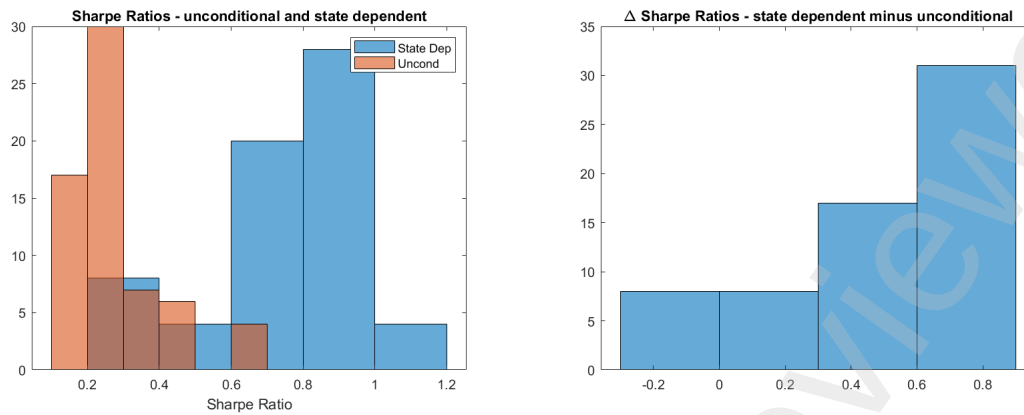


Figure 11: Histograms of unconditional and conditional Sharpe Ratios for the 64 average carry strategies discussed in the text (left panel) and the difference between the two (conditional - unconditional) (right panel). Descriptive analysis: all-sample regression coefficients.

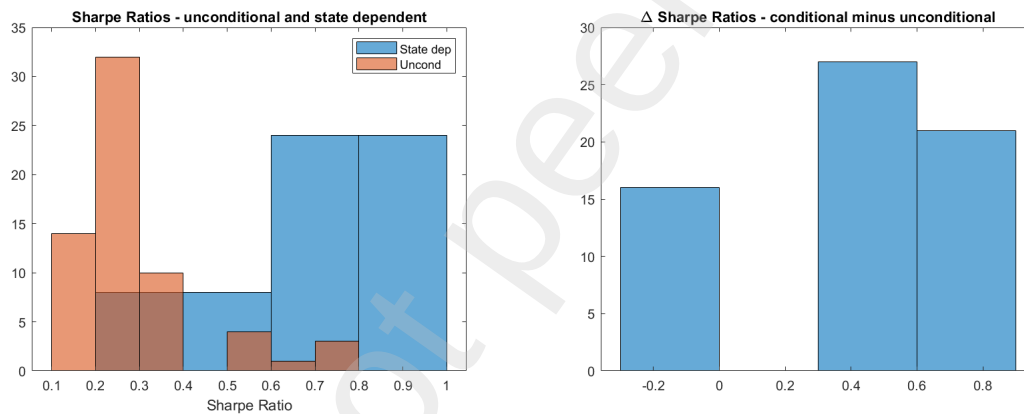


Figure 12: Histograms of unconditional and conditional Sharpe Ratios for the 64 average carry strategies discussed in the text (left panel) and the difference between the two (conditional - unconditional) (right panel). Predictive analysis: regression coefficients from predictive regression with ever-increasing window.

Ratios against the diversity index defined in Section 5.1. Recall that, for the unconditional case we found a negative relationship for the Sharpe ratio, and no relationship for the excess return. When we run the same regressions in Eq 23 and 24 using as left-hand variables the

conditional quantities,

$$\begin{aligned} \langle SR(X, L, H)^{cond} \rangle &= a + b \cdot div(X, L, H) + \epsilon \\ \langle xret(X, L, H)^{cond} \rangle &= A + B \cdot div(X, L, H) + \eta \end{aligned} \quad (31)$$

we now find highly significant slope coefficients, b and B (t statistics 4.36 and 3.98, respectively), R^2 of 0.23 and 0.21, and a *positive* dependence between the diversity index and both the Sharpe Ratio and the excess return. Recall that in the unconditional case we found that the extra volatility brought about by investing and/or funding in a currency different from the investor's more than negated any benefit from having more currencies to choose from. When the investor can scale the notional of each carry strategy based on the expected next-period return, this is no longer the case, and we find that strategies with a high diversity index on average outperform low-diversity strategies both in terms of excess returns and of Sharpe Ratios. This shows that the ability to predict excess returns more than compensates for the extra volatility introduced by the currency mismatch (which in itself would reduce the Sharpe Ratio).

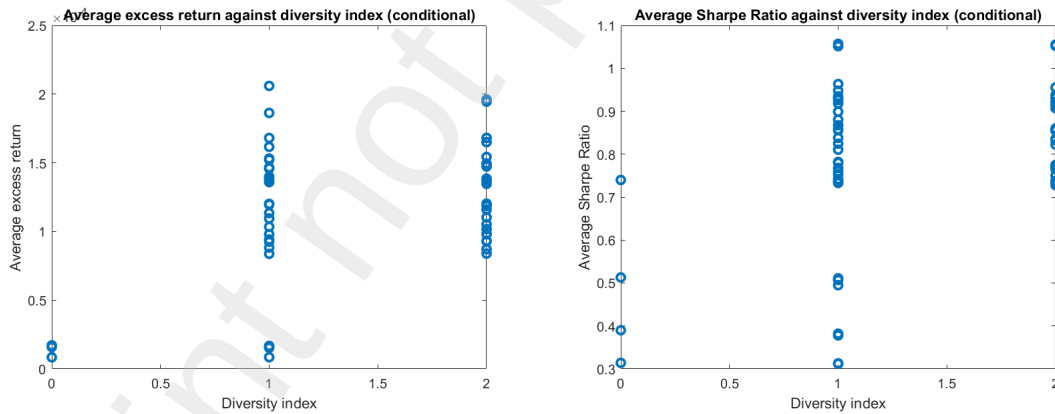


Figure 13: The average excess returns (left panel) and Sharpe Ratios (right panel) for the 64 generalized strategies as a function of the diversity index defined in Section 5.1, for the conditional case.

Finally, we compare the conditional and unconditional average Sharpe Ratios (aver-

HL(X=USD)	HL(X=GBP)	HL(X=EUR)	HL(X=JPN)	<UNC>	<CND>
0.331	0.288	0.334	0.379	0.283	0.765

Table 11: The average Sharpe Ratios for the High-Low strategies when the domestic currency is USD, GBP, EUR and JPN (first four columns), for the unconditional strategy (fifth column), and for the conditional strategy (sixth column).

aged over strategies) and the Sharpe Ratio associated with a common strategy followed by investment professionals, namely, to invest systematically at every point in time in the highest-yielding currency and to fund at the lowest yielding currency. (We call this strategy the ‘High-Low’ strategy.) The results are shown in Tab 11 which presents the average Sharpe Ratios for the High-Low strategies when the domestic currency is USD, GBP, EUR and JPN (first four columns), for the unconditional strategy (fifth column), and for the conditional strategy (sixth column). While the High-Low strategy produces an average Sharpe Ratio that is not statistically different (at the 95th confidence level) from the unconditional strategies, the conditional strategy produces an average Sharpe ratio that is statistically (at the 99th confidence level) and economically different from all the other strategies.

6 Conclusions

In this work we have empirically estimated and theoretically decomposed the various components of the profitability of a generalized carry strategy in which the currency of the investor, of the funding leg and of the investment leg can all be different. This strategy is of theoretical and practical interest, because it sheds light on the economic drivers of the profitability (the various risk premia), and is routinely employed by professional investors. We have expressed the various contributions to the profitability of the strategy in terms of covariances between future/forwards ratios (FX and interest-rate) and the stochastic discount factor in the domestic currency. This has allowed us to provide an economic explanation of what risk exposure the profitability of the strategy offers a compensation for.

We have shown that, as far as the unconditional strategy goes, the extra flexibility afforded by the ability to invest and fund in currencies other than the ‘domestic’ one does not give rise to higher Sharpe ratios, as the extra volatility introduced by the currency mismatch negates the advantages of higher expected returns. Indeed, a regression of Sharpe Ratios against a diversity index that measures the degree of currency mismatch shows a negative slope coefficient.

These conclusions are reversed when we look at what we have called the conditional strategies. We have found that there is predictability in the excess returns produced by the unconditional strategies. Given this predictability, we have built a conditional strategy that scales the constant notional of the unconditional strategy by the expected excess return over the next period. We have found that these conditional strategies have much higher Sharpe Ratios and much lower drawdowns, and that, unlike the unconditional strategies, they benefit substantially from the ability to fund and invest in different currencies. Their superiority over the unconditional strategies is observed both in a descriptive (in-sample) and in a predictive (out-of-sample) analysis. These conditional strategies also display a substantially higher Sharpe Ratio than the ‘High-Low’ strategy commonly employed by market practitioners of investing in the highest-yielding and funding in the lowest-yielding currency.

Finally, we have shown that a generalized strategy in which the investment and funding currency are both different from the domestic currency is not the same as the sum of the associated interest-rate-only and UIP strategies. We have used the powerful non-parametric resampling method by Crump and Gospodinov (2019) to show that the difference is statistically significant for most strategies. The magnitude of this difference depends both in sign and magnitude on the X/H currency pairs, but, as we have shown, for some currency pairs it is economically very significant. Since a covariance term that captures both the interest-rate volatilities and the correlations between domestic and investment

interest rates is at the heart of these difference, we suspect that it could be significantly larger when the investment is made in the government bonds of an emerging currency. We have been prevented from carrying out this analysis by the dearth of the reliable, high-quality data required for the task, but, if such data were to become available, this would be an obvious direction in which our research could be extended.

A Bootstrap Resampling Method and Test

The Crump and Gospodinov (2019) resampling technique is built from time series of excess returns, which are good building blocks for the term structure because they retain the time series properties of returns; they do not contain any overlapping maturity periods in their construction; and they have a transparent economic interpretation. The method has been shown to be asymptotically valid and demonstrates excellent finite-sample properties in simulations, making it particularly well-suited for our analysis.

In this appendix we briefly review the method, and presents those tests of its performance that are of greatest relevance for our study.

A.1 The Setting

Let p_t^n denote the time t log price of a zero-coupon bond with n periods to maturity which pay 1 unit at time $t + n$ where $t = 1, \dots, N$. The corresponding log yield is denoted by y_t^n and satisfies $p_t^n = -n \cdot y_t^n$. The log forward rate corresponding to a one-period investment between $t + n - 1$ and $t + n$, f_t^n , is defined as:

$$f_t^n = p_t^{n-1} - p_t^n \quad (32)$$

$p_t^0 = 0$, the one-period rate, y_t^1 , is equal to f_t^1 . Using the recursive nature of Eq 32 and the definition of yields, we have

$$p_t^n = -\sum_{i=1}^n f_t^i, \quad y_t^n = \frac{1}{n} \sum_{i=1}^n f_t^i \quad (33)$$

Define the one-period holding return on a bond of maturity n from time t to $t + 1$ as

$$r_{t,t+1}^n = p_{t+1}^{n-1} - p_t^n \quad (34)$$

Then the difference return can be written as

$$dr_{t,t+1}^n = (p_{t+1}^{n-1} - p_t^n) - (p_{t+1}^{n-2} - p_t^{n-1}) = r_{t+1}^n - r_{t+1}^{n-1} = f_t^n - f_{t+1}^{n-1} \quad (35)$$

A.2 Bootstrap

The bootstrapping procedure proceeds as follows. We start from given spot yields y_t^n and calculate forward yields f_t^i using Eq.33. For the maximum maturity (N), we calculate the difference in forward rates as

$$\Delta f_{t+1}^N = f_{t+1}^N - f_t^N \quad (36)$$

To synchronize the resampled foreign exchange rate simultaneously, we calculate the difference in log foreign exchange as

$$\Delta LFX_{t+1} = \log(FX_{t+1}) - \log(FX_t) \quad (37)$$

Next we stack the quantities $f_{t+1}^N, dr_{t+1}^{n+1}$, and ΔLFX_{t+1} in the matrix Z as given by

$$Z = \begin{bmatrix} \Delta f_2^N & \Delta r_2^2 & \dots & dr_2^N \\ \Delta f_3^N & \Delta r_3^2 & \dots & dr_3^N \\ \dots & \dots & \dots & \dots \\ \Delta f_T^N & \Delta r_T^2 & \dots & dr_T^N \end{bmatrix} \quad (38)$$

where T is the number of observation dates in the dataset. In our research, the T is 600 and each row of Z includes 39 elements - 36 forward rates (1-9 maturities for 4 countries) and 3 FX rates (EUR/USD, GBP/USD, and JPN/USD). The bootstrap samples are then obtained by drawing with replacement k blocks of $M(1 < M < T)$ observations from matrix Z . k is the nearest integer smaller than or equal to $(T - 1)/M$. Using methods of Politis and White (2004), whose algorithm is designed to get the minimum size needed to preserve serial correlation throughout the data points, we estimate the minimum block length M as 45-48 for forwards and 44-52 for FX. Besides, Crump and Gospodinov (2019) proposals a simple rule-of-thumb choice for M : $M = (T \times \text{number of columns})^{2/5}$, which is 55.94 for our data. Therefore, we select a block length of 60, the nearest larger multiple of ten.

More precisely, given a bootstrap sample $\{\Delta f_t^N, dr_t^2, \dots, dr_t^N\}$, the bootstrapped forward

rates and FX rates are constructed, using Eq.35,36, and 37, as

$$\begin{aligned}
 FX_{t+1} &= FX_t \cdot e^{\Delta LFX_{t+1}} \\
 f_{t+1}^N &= f_t^N + \Delta f_{t+1}^N \\
 f_{t+1}^{N-1} &= f_t^{N-1} - dr_{t+1}^N \\
 f_{t+1}^{N-2} &= f_t^{N-1} - dr_{t+1}^{N-1} \\
 &\dots \\
 f_{t+1}^1 &= f_t^2 - dr_{t+1}^2
 \end{aligned} \tag{39}$$

Finally, given a random initial forward curve and k bootstrap samples, we can get one synchronized curve of forward rates and FX rates. We did this 1000 times and calculated the average as the final synchronized curve. The bootstrap yields and bond prices are obtained using Eq.33.

A.3 Test of Resampling

After resampling, it is necessary to test whether important properties of the resampled curves match the corresponding original empirical quantities. If so, it is rational to use this resampling method for robust study. We did this by comparing the series auto-correlation, cross-sectional correlations, eigenvalues of correlations, and the volatility for resampling curves with those characteristics for original empirical yield and FX curves. Fig 15 (Fig 14 presents a comparison of the serial correlation of the first-order differences in 9-y (1-y) maturity yields between the resampled results and the original values for the US, EU, UK, and Japan, respectively. We can see that the resampled yield curves have nearly the same autocorrelation properties of the original yield curves. For FX rates, the top part of Fig 16 presents a comparison of the serial correlation of the first-order differences in FX rate between the resampled results and the original values for the EUR/USD, GBP/USD, and JPN/USD, respectively. The bottom part of Fig 16 compares the cross-sectional correlations of the FX rate with all the 39 variables (1-9 year maturity yields for four countries and the three FX rates) between the resampled results and the original values for EUR/USD,

GBP/USD, and JPN/USD. The results show that, despite slight errors, the resampled FX curves effectively preserve the autocorrelation and cross-sectional correlation properties of the original FX curves. Besides, the left part of Fig 17 compares the eigenvalues of the correlation matrix for all 39 variables between the resampled results and the original values. The right part of the figure compares the volatility of all 39 variables. The results show that the resampled data closely mimic the original values. Therefore, it is rational to use this resampling method in our study because it can preserve the properties of original curves such as serial correlation and cross-sectional relationships.

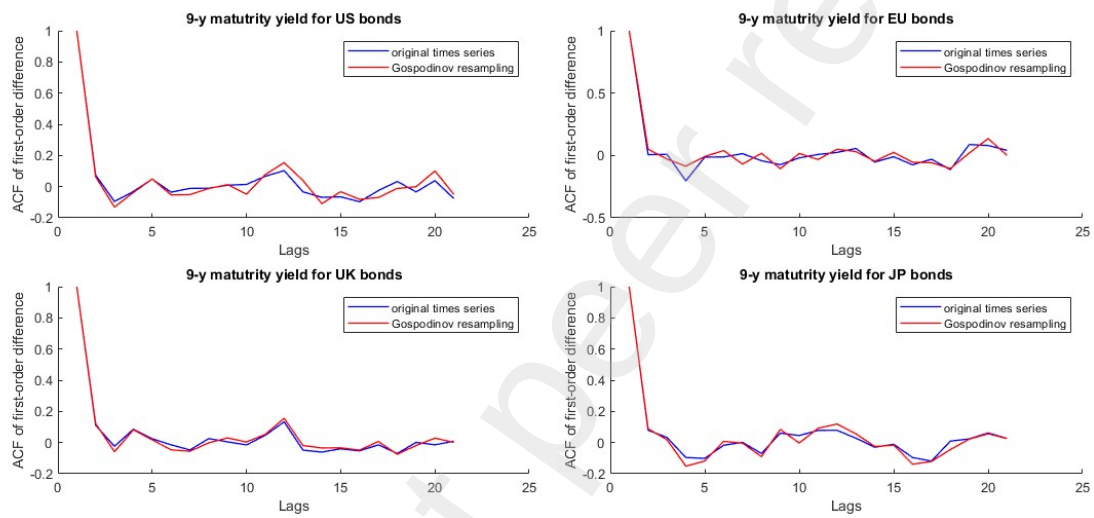


Figure 14: **Comparison of ACF for Δ yield (9-y maturity) between resampling and original curve.** This figure presents a comparison of the serial auto-correlation (ACF) of the first-order differences in 9-y maturity yields with lag 21 between the resampled results and the original values for the US, EU, UK, and Japan, respectively.

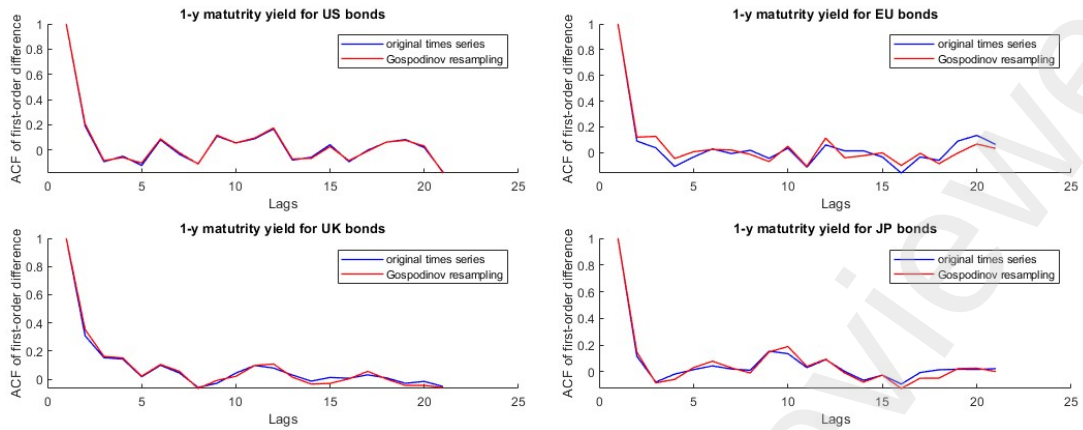


Figure 15: **Comparison of ACF for Δ yield (1-y maturity) between resampling and original curve.** This figure presents a comparison of the serial auto-correlation (ACF) of the first-order differences in 1-y maturity yields with lag 21 between the resampled results and the original values for the US, EU, UK, and Japan, respectively.

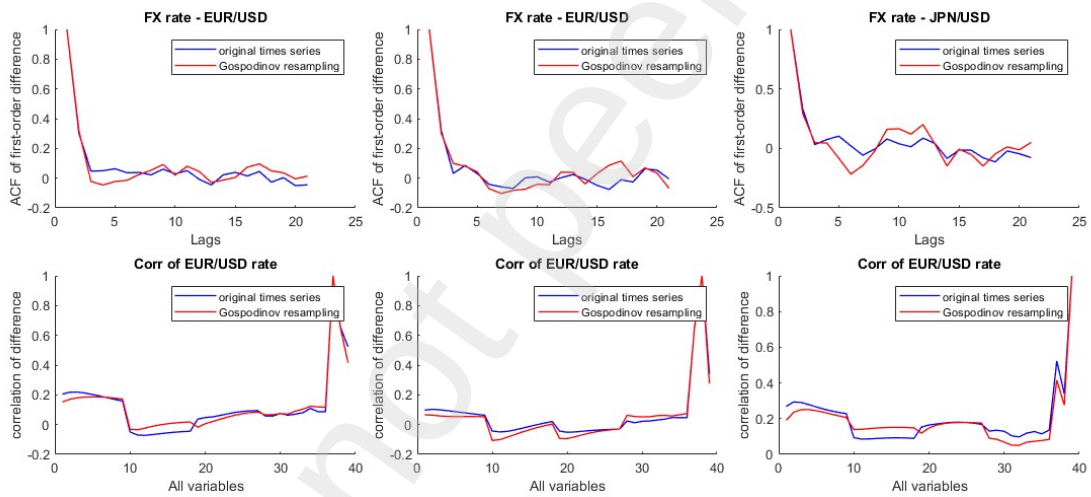


Figure 16: **Comparison of ACF for Δ FX rates and correlation of FX rates with other variables between resampling and original curve.** In this figure, the top part presents a comparison of the serial correlation of the first-order differences in FX rate with lag 21 between the resampled results and the original values for the EUR/USD, GBP/USD, and JPN/USD, respectively. The bottom part compares the cross-sectional correlations of the FX rate with all the 39 variables (1-9 year maturity yields for four countries and the three FX rates) between the resampled results and the original values for EUR/USD, GBP/USD, and JPN/USD.

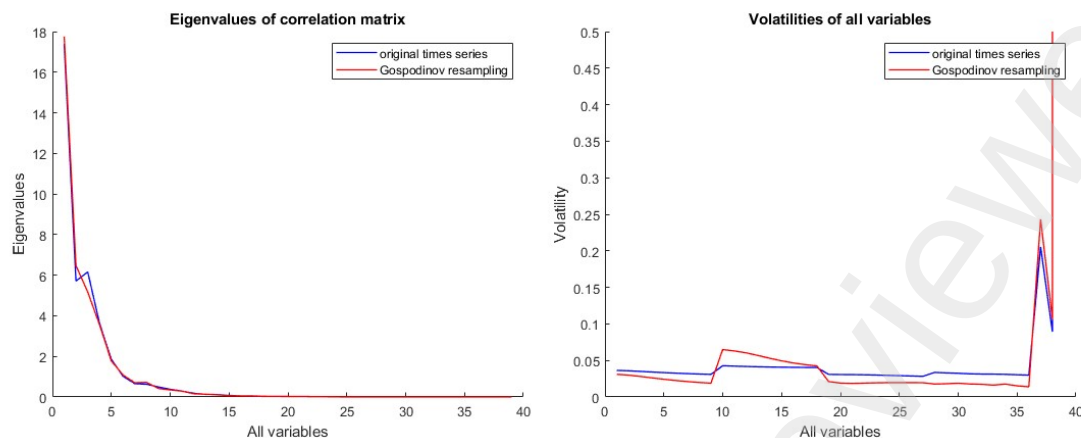


Figure 17: **Comparison of eigenvalues of correlation matrix and volatility between resampling and original curve.** In this figure, the left part compares the eigenvalues of the correlation matrix for all 39 variables between the resampled results and the original values. The right part of the figure compares the volatility of all 39 variables between the resampled results and the original values.

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